

Thermodynamics

Instructor

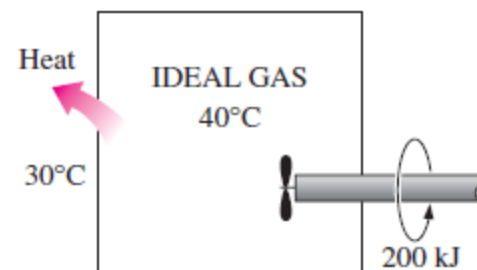
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Chapter 7 – Problems

**Problem numbers correspond to
5th Edition**

Entropy and the Increase of Entropy Principle

7-24 A rigid tank contains an ideal gas at 40°C that is being stirred by a paddle wheel. The paddle wheel does 200 kJ of work on the ideal gas. It is observed that the temperature of the ideal gas remains constant during this process as a result of heat transfer between the system and the surroundings at 30°C. Determine the entropy change of the ideal gas.

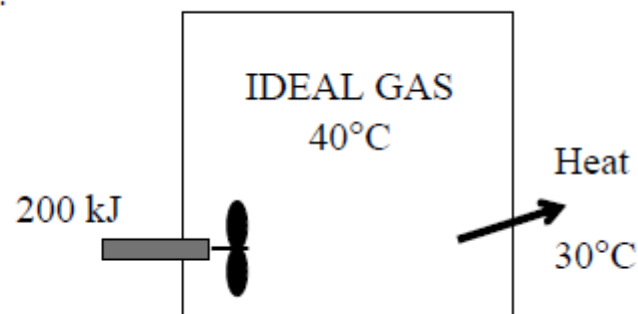


Assumptions The gas in the tank is given to be an ideal gas.

Analysis The temperature and the specific volume of the gas remain constant during this process.

Therefore, the initial and the final states of the gas are the same. Then $s_2 = s_1$ since entropy is a property. Therefore,

$$\Delta S_{\text{sys}} = 0$$



7-25 Air is compressed by a 12-kW compressor from P_1 to P_2 . The air temperature is maintained constant at 25°C during this process as a result of heat transfer to the surrounding medium at 10°C . Determine the rate of entropy change of the air. State the assumptions made in solving this problem.

Answer: -0.0403 kW/K

Properties Noting that $h = h(T)$ for ideal gases, we have $h_1 = h_2$ since $T_1 = T_2 = 25^\circ\text{C}$.

Analysis We take the compressor as the system. Noting that the enthalpy of air remains constant, the energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

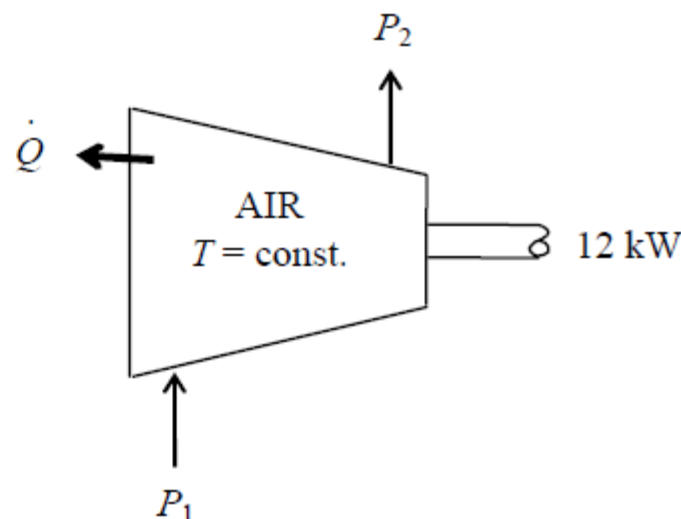
$$\dot{W}_{\text{in}} = \dot{Q}_{\text{out}}$$

Therefore,

$$\dot{Q}_{\text{out}} = \dot{W}_{\text{in}} = 12 \text{ kW}$$

Noting that the process is assumed to be an isothermal and internally reversible process, the rate of entropy change of air is determined to be

$$\Delta \dot{S}_{\text{air}} = -\frac{\dot{Q}_{\text{out,air}}}{T_{\text{sys}}} = -\frac{12 \text{ kW}}{298 \text{ K}} = -0.0403 \text{ kW/K}$$



7-29 Refrigerant-134a enters the coils of the evaporator of a refrigeration system as a saturated liquid–vapor mixture at a pressure of 160 kPa. The refrigerant absorbs 180 kJ of heat from the cooled space, which is maintained at -5°C , and leaves as saturated vapor at the same pressure. Determine (a) the entropy change of the refrigerant, (b) the entropy change of the cooled space, and (c) the total entropy change for this process.

Analysis Noting that both the refrigerant and the cooled space undergo reversible isothermal processes, the entropy change for them can be determined from

$$\Delta S = \frac{Q}{T}$$

(a) The pressure of the refrigerant is maintained constant. Therefore, the temperature of the refrigerant also remains constant at the saturation value,

$$T = T_{\text{sat}@160 \text{ kPa}} = -15.6^{\circ}\text{C} = 257.4 \text{ K} \quad (\text{Table A-12})$$

Then,

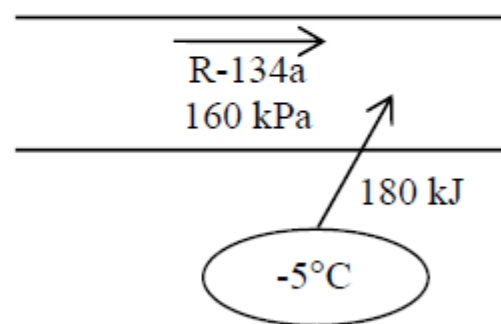
$$\Delta S_{\text{refrigerant}} = \frac{Q_{\text{refrigerant,in}}}{T_{\text{refrigerant}}} = \frac{180 \text{ kJ}}{257.4 \text{ K}} = \mathbf{0.699 \text{ kJ/K}}$$

(b) Similarly,

$$\Delta S_{\text{space}} = -\frac{Q_{\text{space,out}}}{T_{\text{space}}} = -\frac{180 \text{ kJ}}{268 \text{ K}} = \mathbf{-0.672 \text{ kJ/K}}$$

(c) The total entropy change of the process is

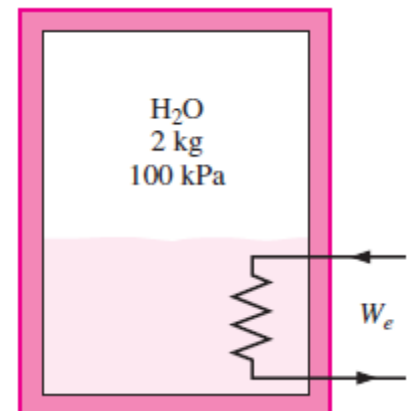
$$S_{\text{gen}} = S_{\text{total}} = \Delta S_{\text{refrigerant}} + \Delta S_{\text{space}} = 0.699 - 0.672 = \mathbf{0.027 \text{ kJ/K}}$$



Entropy Changes of Pure Substances

7-34 A well-insulated rigid tank contains 2 kg of a saturated liquid–vapor mixture of water at 100 kPa. Initially, three-quarters of the mass is in the liquid phase. An electric resistance heater placed in the tank is now turned on and kept on until all the liquid in the tank is vaporized. Determine the entropy change of the steam during this process. *Answer:*

8.10 kJ/K



7-35



A rigid tank is divided into two equal parts by a partition. One part of the tank contains 1.5 kg of compressed liquid water at 300 kPa and 60°C while the other part is evacuated. The partition is now removed, and the water expands to fill the entire tank. Determine the entropy change of water during this process, if the final pressure in the tank is 15 kPa. *Answer: -0.114 kJ/K*

1.5 kg compressed liquid 300 kPa 60°C	Vacuum
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Analysis The properties of the water are (Table A-4)

$$\left. \begin{array}{l} P_1 = 300 \text{ kPa} \\ T_1 = 60^\circ\text{C} \end{array} \right\} \begin{array}{l} \nu_1 \cong \nu_{f@60^\circ\text{C}} = 0.001017 \text{ m}^3/\text{kg} \\ s_1 = s_{f@60^\circ\text{C}} = 0.8313 \text{ kJ/kg} \cdot \text{K} \end{array}$$

Noting that

$$\nu_2 = 2\nu_1 = (2)(0.001017) = 0.002034 \text{ m}^3/\text{kg}$$

$$\left. \begin{array}{l} P_2 = 15 \text{ kPa} \\ \nu_2 = 0.002034 \text{ m}^3/\text{kg} \end{array} \right\} \begin{array}{l} x_2 = \frac{\nu_2 - \nu_f}{\nu_{fg}} = \frac{0.002034 - 0.001014}{10.02 - 0.001014} = 0.0001018 \\ s_2 = s_f + x_2 s_{fg} = 0.7549 + (0.0001018)(7.2522) = 0.7556 \text{ kJ/kg} \cdot \text{K} \end{array}$$

Then the entropy change of the water becomes

$$\Delta S = m(s_2 - s_1) = (1.5 \text{ kg})(0.7556 - 0.8313) \text{ kJ/kg} \cdot \text{K} = \mathbf{-0.114 \text{ kJ/K}}$$

1.5 kg compressed liquid 300 kPa 60°C	Vacuum
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7-38 An insulated piston-cylinder device contains 5 L of saturated liquid water at a constant pressure of 150 kPa. An electric resistance heater inside the cylinder is now turned on, and 2200 kJ of energy is transferred to the steam. Determine the entropy change of the water during this process.

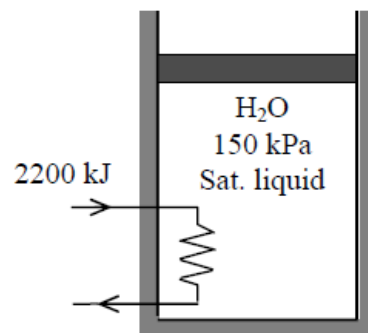
Answer: 5.72 kJ/K

Analysis From the steam tables (Tables A-4 through A-6),

$$P_1 = 150 \text{ kPa} \left\{ \begin{array}{l} \nu_1 = \nu_{f@150 \text{ kPa}} = 0.001053 \text{ m}^3/\text{kg} \\ h_1 = h_{f@150 \text{ kPa}} = 467.13 \text{ kJ/kg} \\ s_1 = s_{f@150 \text{ kPa}} = 1.4337 \text{ kJ/kg} \cdot \text{K} \end{array} \right.$$

Also,

$$m = \frac{\nu}{\nu_1} = \frac{0.005 \text{ m}^3}{0.001053 \text{ m}^3/\text{kg}} = 4.75 \text{ kg}$$



We take the contents of the cylinder as the system. This is a closed system since no mass enters or leaves. The energy balance for this stationary closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$W_{\text{e,in}} - W_{\text{b,out}} = \Delta U$$

$$W_{\text{e,in}} = m(h_2 - h_1)$$

since $\Delta U + W_{\text{b}} = \Delta H$ during a constant pressure quasi-equilibrium process. Solving for h_2 ,

$$h_2 = h_1 + \frac{W_{\text{e,in}}}{m} = 467.13 + \frac{2200 \text{ kJ}}{4.75 \text{ kg}} = 930.33 \text{ kJ/kg}$$

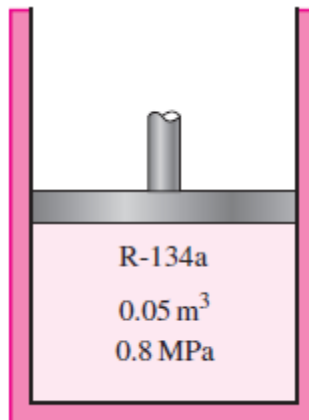
Thus,

$$\left. \begin{array}{l} P_2 = 150 \text{ kPa} \\ h_2 = 930.33 \text{ kJ/kg} \end{array} \right\} \begin{array}{l} x_2 = \frac{h_2 - h_f}{h_{fg}} = \frac{930.33 - 467.13}{2226.0} = 0.2081 \\ s_2 = s_f + x_2 s_{fg} = 1.4337 + (0.2081)(5.7894) = 2.6384 \text{ kJ/kg} \cdot \text{K} \end{array}$$

Then the entropy change of the water becomes

$$\Delta S = m(s_2 - s_1) = (4.75 \text{ kg})(2.6384 - 1.4337) \text{ kJ/kg} \cdot \text{K} = 5.72 \text{ kJ/K}$$

7-39 An insulated piston–cylinder device contains 0.05 m^3 of saturated refrigerant-134a vapor at 0.8-MPa pressure. The refrigerant is now allowed to expand in a reversible manner until the pressure drops to 0.4 MPa . Determine (a) the final temperature in the cylinder and (b) the work done by the refrigerant.



7-41 Refrigerant-134a enters an adiabatic compressor as saturated vapor at 160 kPa at a rate of 2 m³/min and is compressed to a pressure of 900 kPa. Determine the minimum power that must be supplied to the compressor.

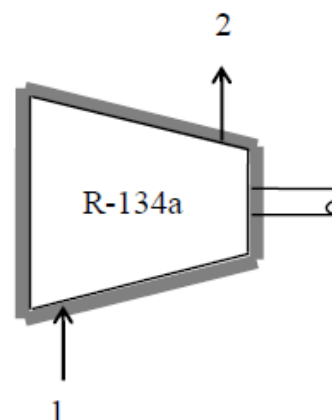
Analysis The power input to an adiabatic compressor will be a minimum when the compression process is reversible. For the reversible adiabatic process we have $s_2 = s_1$. From the refrigerant tables (Tables A-11 through A-13),

$$\left. \begin{array}{l} P_1 = 160 \text{ kPa} \\ \text{sat. vapor} \end{array} \right\} \begin{array}{l} \nu_1 = \nu_{g@160 \text{ kPa}} = 0.12348 \text{ m}^3/\text{kg} \\ h_1 = h_{g@160 \text{ kPa}} = 241.11 \text{ kJ/kg} \\ s_1 = s_{g@160 \text{ kPa}} = 0.9419 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_2 = 900 \text{ kPa} \\ s_2 = s_1 \end{array} \right\} h_2 = 277.06 \text{ kJ/kg}$$

Also,

$$\dot{m} = \frac{\dot{V}_1}{\nu_1} = \frac{2 \text{ m}^3/\text{min}}{0.12348 \text{ m}^3/\text{kg}} = 16.20 \text{ kg/min} = 0.27 \text{ kg/s}$$



There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. We take the compressor as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \overset{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{W}_{\text{in}} + \dot{m}h_1 = \dot{m}h_2 \quad (\text{since } \dot{Q} \equiv \Delta \dot{ke} \equiv \Delta \dot{pe} \equiv 0)$$

$$\dot{W}_{\text{in}} = \dot{m}(h_2 - h_1)$$

Substituting, the minimum power supplied to the compressor is determined to be

$$\dot{W}_{\text{in}} = (0.27 \text{ kg/s})(277.06 - 241.11) \text{ kJ/kg} = \mathbf{9.71 \text{ kW}}$$

7-44 A heavily insulated piston–cylinder device contains 0.05 m^3 of steam at 300 kPa and 150°C . Steam is now compressed in a reversible manner to a pressure of 1 MPa . Determine the work done on the steam during this process.

7-46 A piston–cylinder device contains 1.2 kg of saturated water vapor at 200°C. Heat is now transferred to steam, and steam expands reversibly and isothermally to a final pressure of 800 kPa. Determine the heat transferred and the work done during this process.

Analysis From the steam tables (Tables A-4 through A-6),

$$\left. \begin{array}{l} T_1 = 200^\circ\text{C} \\ \text{sat. vapor} \end{array} \right\} \begin{array}{l} u_1 = u_{g@200^\circ\text{C}} = 2594.2 \text{ kJ/kg} \\ s_1 = s_{g@200^\circ\text{C}} = 6.4302 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_2 = 800 \text{ kPa} \\ T_2 = T_1 \end{array} \right\} \begin{array}{l} u_2 = 2631.1 \text{ kJ/kg} \\ s_2 = 6.8177 \text{ kJ/kg} \cdot \text{K} \end{array}$$

The heat transfer for this reversible isothermal process can be determined from

$$Q = T\Delta S = Tm(s_2 - s_1) = (473 \text{ K})(1.2 \text{ kg})(6.8177 - 6.4302) \text{ kJ/kg} \cdot \text{K} = \mathbf{219.9 \text{ kJ}}$$

We take the contents of the cylinder as the system. This is a closed system since no mass enters or leaves. The energy balance for this closed system can be expressed as

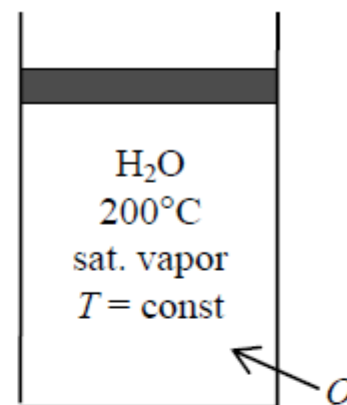
$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$Q_{\text{in}} - W_{\text{b,out}} = \Delta U = m(u_2 - u_1)$$

$$W_{\text{b,out}} = Q_{\text{in}} - m(u_2 - u_1)$$

Substituting, the work done during this process is determined to be

$$W_{\text{b,out}} = 219.9 \text{ kJ} - (1.2 \text{ kg})(2631.1 - 2594.2) \text{ kJ/kg} = \mathbf{175.6 \text{ kJ}}$$



7-48 A piston–cylinder device contains 5 kg of steam at 100°C with a quality of 50 percent. This steam undergoes two processes as follows:

1-2 Heat is transferred to the steam in a reversible manner while the temperature is held constant until the steam exists as a saturated vapor.

2-3 The steam expands in an adiabatic, reversible process until the pressure is 15 kPa.

(a) Sketch these processes with respect to the saturation lines on a single T - s diagram.

(b) Determine the heat added to the steam in process 1-2, in kJ.

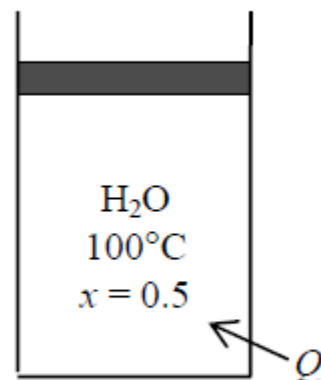
(c) Determine the work done by the steam in process 2-3, in kJ.

Analysis (b) From the steam tables (Tables A-4 through A-6),

$$\left. \begin{array}{l} T_1 = 100^\circ\text{C} \\ x = 0.5 \end{array} \right\} h_1 = h_f + xh_{fg} = 419.17 + (0.5)(2256.4) = 1547.4 \text{ kJ/kg}$$

$$\left. \begin{array}{l} T_2 = 100^\circ\text{C} \\ x_2 = 1 \end{array} \right\} \begin{array}{l} h_2 = h_g = 2675.6 \text{ kJ/kg} \\ u_2 = u_g = 2506.0 \text{ kJ/kg} \\ s_2 = 7.3542 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_3 = 15 \text{ kPa} \\ s_3 = s_2 \end{array} \right\} u_3 = 2247.9 \text{ kJ/kg}$$



We take the contents of the cylinder as the system. This is a closed system since no mass enters or leaves. The energy balance for this closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

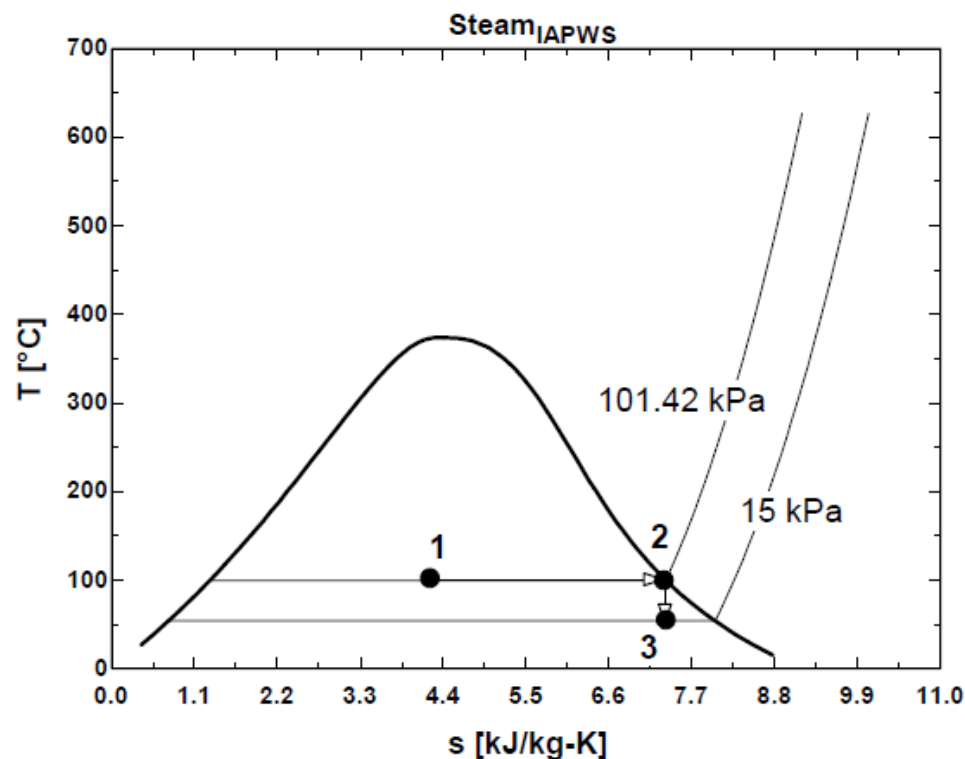
$$Q_{\text{in}} - W_{\text{b,out}} = \Delta U = m(u_2 - u_1)$$

For process 1-2, it reduces to

$$Q_{12,\text{in}} = m(h_2 - h_1) = (5 \text{ kg})(2675.6 - 1547.4) \text{ kJ/kg} = \mathbf{5641 \text{ kJ}}$$

(c) For process 2-3, it reduces to

$$W_{23,\text{b,out}} = m(u_2 - u_3) = (5 \text{ kg})(2506.0 - 2247.9) \text{ kJ/kg} = \mathbf{1291 \text{ kJ}}$$



Entropy Change of Incompressible Substances

7-53 A 50-kg copper block initially at 80°C is dropped into an insulated tank that contains 120 L of water at 25°C. Determine the final equilibrium temperature and the total entropy change for this process.

Analysis We take the entire contents of the tank, water + copper block, as the *system*. This is a *closed system* since no mass crosses the system boundary during the process. The energy balance for this system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{\text{system}}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc. energies}}}$$

$$0 = \Delta U$$

or,

$$\Delta U_{\text{Cu}} + \Delta U_{\text{water}} = 0$$

$$[mc(T_2 - T_1)]_{\text{Cu}} + [mc(T_2 - T_1)]_{\text{water}} = 0$$

where

$$m_{\text{water}} = \rho V = (997 \text{ kg/m}^3)(0.120 \text{ m}^3) = 119.6 \text{ kg}$$

Using specific heat values for copper and liquid water at room temperature and substituting,

$$(50 \text{ kg})(0.386 \text{ kJ/kg} \cdot ^\circ\text{C})(T_2 - 80)^\circ\text{C} + (119.6 \text{ kg})(4.18 \text{ kJ/kg} \cdot ^\circ\text{C})(T_2 - 25)^\circ\text{C} = 0$$

$$T_2 = 27.0^\circ\text{C}$$

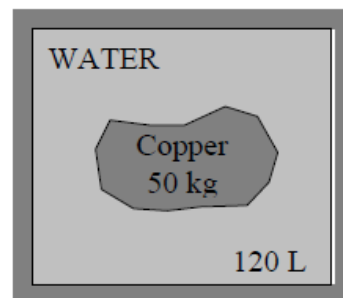
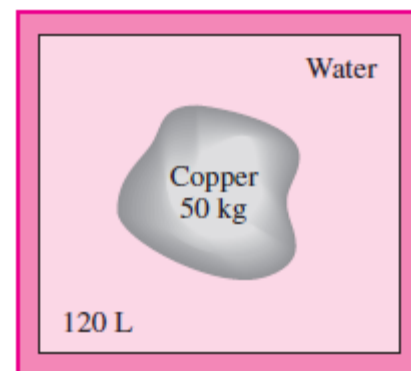
The entropy generated during this process is determined from

$$\Delta S_{\text{copper}} = mc_{\text{avg}} \ln \left(\frac{T_2}{T_1} \right) = (50 \text{ kg})(0.386 \text{ kJ/kg} \cdot \text{K}) \ln \left(\frac{300.0 \text{ K}}{353 \text{ K}} \right) = -3.140 \text{ kJ/K}$$

$$\Delta S_{\text{water}} = mc_{\text{avg}} \ln \left(\frac{T_2}{T_1} \right) = (119.6 \text{ kg})(4.18 \text{ kJ/kg} \cdot \text{K}) \ln \left(\frac{300.0 \text{ K}}{298 \text{ K}} \right) = 3.344 \text{ kJ/K}$$

Thus,

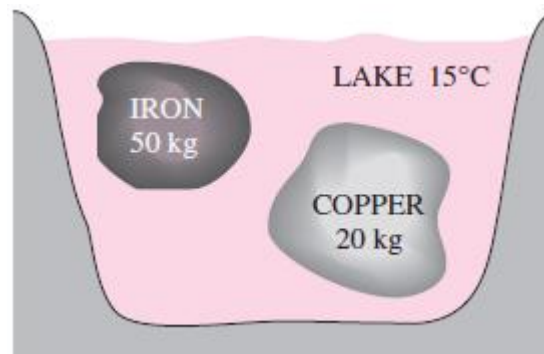
$$\Delta S_{\text{total}} = \Delta S_{\text{copper}} + \Delta S_{\text{water}} = -3.140 + 3.344 = \mathbf{0.204 \text{ kJ/K}}$$



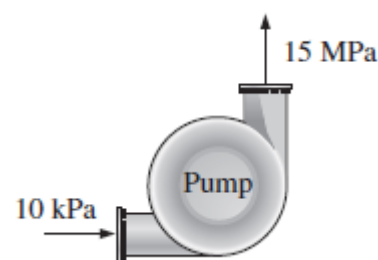
7-54 A 25-kg iron block initially at 350°C is quenched in an insulated tank that contains 100 kg of water at 18°C . Assuming the water that vaporizes during the process condenses back in the tank, determine the total entropy change during this process.

7-55 A 20-kg aluminum block initially at 200°C is brought into contact with a 20-kg block of iron at 100°C in an insulated enclosure. Determine the final equilibrium temperature and the total entropy change for this process. *Answers:* 168.4°C , 0.169 kJ/K

7-57 A 50-kg iron block and a 20-kg copper block, both initially at 80°C , are dropped into a large lake at 15°C . Thermal equilibrium is established after a while as a result of heat transfer between the blocks and the lake water. Determine the total entropy change for this process.



7-58 An adiabatic pump is to be used to compress saturated liquid water at 10 kPa to a pressure to 15 MPa in a reversible manner. Determine the work input using (a) entropy data from the compressed liquid table, (b) inlet specific volume and pressure values, (c) average specific volume and pressure values. Also, determine the errors involved in parts (b) and (c).



Assumptions 1 Steady operating conditions exist. 2 Kinetic and potential energy changes are negligible. 3 Heat transfer to or from the fluid is negligible.

Analysis The properties of water at the inlet and exit of the pump are (Tables A-4 through A-6)

$$\left. \begin{array}{l} P_1 = 10 \text{ kPa} \\ x_1 = 0 \end{array} \right\} \begin{array}{l} h_1 = 191.81 \text{ kJ/kg} \\ s_1 = 0.6492 \text{ kJ/kg} \\ \nu_1 = 0.001010 \text{ m}^3/\text{kg} \end{array}$$

$$\left. \begin{array}{l} P_2 = 15 \text{ MPa} \\ s_2 = s_1 \end{array} \right\} \begin{array}{l} h_2 = 206.90 \text{ kJ/kg} \\ \nu_2 = 0.001004 \text{ m}^3/\text{kg} \end{array}$$

(a) Using the entropy data from the compressed liquid water table

$$w_p = h_2 - h_1 = 206.90 - 191.81 = \mathbf{15.10 \text{ kJ/kg}}$$

(b) Using inlet specific volume and pressure values

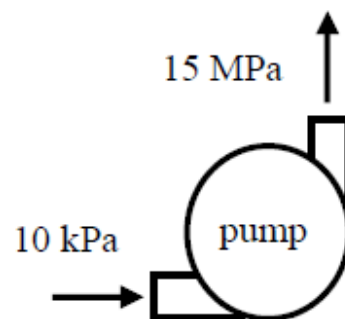
$$w_p = \nu_1 (P_2 - P_1) = (0.001010 \text{ m}^3/\text{kg})(15,000 - 10) \text{ kPa} = \mathbf{15.14 \text{ kJ/kg}}$$

$$\text{Error} = \mathbf{0.3\%}$$

(c) Using average specific volume and pressure values

$$w_p = \nu_{\text{avg}} (P_2 - P_1) = \left[1/2(0.001010 + 0.001004) \text{ m}^3/\text{kg} \right] (15,000 - 10) \text{ kPa} = \mathbf{15.10 \text{ kJ/kg}}$$

$$\text{Error} = \mathbf{0\%}$$



Discussion The results show that any of the method may be used to calculate reversible pump work.

7-68 An insulated piston–cylinder device initially contains 300 L of air at 120 kPa and 17°C. Air is now heated for 15 min by a 200-W resistance heater placed inside the cylinder. The pressure of air is maintained constant during this process. Determine the entropy change of air, assuming (a) constant specific heats and (b) variable specific heats.

Properties The gas constant of air is $R = 0.287 \text{ kJ/kg}\cdot\text{K}$ (Table A-1).

Analysis The mass of the air and the electrical work done during this process are

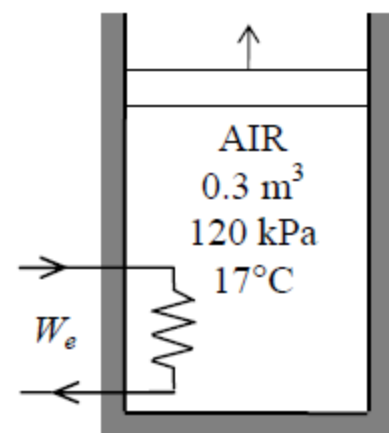
$$m = \frac{P_1 V_1}{RT_1} = \frac{(120 \text{ kPa})(0.3 \text{ m}^3)}{(0.287 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(290 \text{ K})} = 0.4325 \text{ kg}$$

$$W_{e,\text{in}} = \dot{W}_{e,\text{in}} \Delta t = (0.2 \text{ kJ/s})(15 \times 60 \text{ s}) = 180 \text{ kJ}$$

The energy balance for this stationary closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$W_{e,\text{in}} - W_{b,\text{out}} = \Delta U \longrightarrow W_{e,\text{in}} = m(h_2 - h_1) \cong c_p(T_2 - T_1)$$



since $\Delta U + W_b = \Delta H$ during a constant pressure quasi-equilibrium process.

(a) Using a constant c_p value at the anticipated average temperature of 450 K, the final temperature becomes

$$\text{Thus, } T_2 = T_1 + \frac{W_{e,\text{in}}}{mc_p} = 290 \text{ K} + \frac{180 \text{ kJ}}{(0.4325 \text{ kg})(1.02 \text{ kJ/kg}\cdot\text{K})} = 698 \text{ K}$$

Then the entropy change becomes

$$\begin{aligned}\Delta S_{\text{sys}} &= m(s_2 - s_1) = m \left(c_{p,\text{avg}} \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \right) = mc_{p,\text{avg}} \ln \frac{T_2}{T_1} \\ &= (0.4325 \text{ kg})(1.020 \text{ kJ/kg} \cdot \text{K}) \ln \left(\frac{698 \text{ K}}{290 \text{ K}} \right) = \mathbf{0.387 \text{ kJ/K}}\end{aligned}$$

(b) Assuming variable specific heats,

$$W_{\text{e,in}} = m(h_2 - h_1) \longrightarrow h_2 = h_1 + \frac{W_{\text{e,in}}}{m} = 290.16 \text{ kJ/kg} + \frac{180 \text{ kJ}}{0.4325 \text{ kg}} = 706.34 \text{ kJ/kg}$$

From the air table (Table A-17, we read $s_2^\circ = 2.5628 \text{ kJ/kg} \cdot \text{K}$ corresponding to this h_2 value. Then,

$$\Delta S_{\text{sys}} = m \left(s_2^\circ - s_1^\circ + R \ln \frac{P_2}{P_1} \right) = m(s_2^\circ - s_1^\circ) = (0.4325 \text{ kg})(2.5628 - 1.66802) \text{ kJ/kg} \cdot \text{K} = \mathbf{0.387 \text{ kJ/K}}$$

7-69 A piston–cylinder device contains 1.2 kg of nitrogen gas at 120 kPa and 27°C. The gas is now compressed slowly in a polytropic process during which $PV^{1.3} = \text{constant}$. The process ends when the volume is reduced by one-half. Determine the entropy change of nitrogen during this process.

Answer: -0.0617 kJ/K

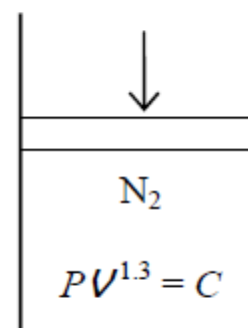
Properties The gas constant of nitrogen is $R = 0.297 \text{ kJ/kg}\cdot\text{K}$ (Table A-1). The constant volume specific heat of nitrogen at room temperature is $c_v = 0.743 \text{ kJ/kg}\cdot\text{K}$ (Table A-2).

Analysis From the polytropic relation,

$$\frac{T_2}{T_1} = \left(\frac{v_1}{v_2} \right)^{n-1} \longrightarrow T_2 = T_1 \left(\frac{v_1}{v_2} \right)^{n-1} = (300 \text{ K})(2)^{1.3-1} = 369.3 \text{ K}$$

Then the entropy change of nitrogen becomes

$$\begin{aligned} \Delta S_{N_2} &= m \left(c_{v,\text{avg}} \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1} \right) \\ &= (1.2 \text{ kg}) \left((0.743 \text{ kJ/kg}\cdot\text{K}) \ln \frac{369.3 \text{ K}}{300 \text{ K}} + (0.297 \text{ kJ/kg}\cdot\text{K}) \ln(0.5) \right) = -0.0617 \text{ kJ/K} \end{aligned}$$



7-74 An insulated rigid tank is divided into two equal parts by a partition. Initially, one part contains 5 kmol of an ideal gas at 250 kPa and 40°C, and the other side is evacuated. The partition is now removed, and the gas fills the entire tank. Determine the total entropy change during this process.

Answer: 28.81 kJ/K

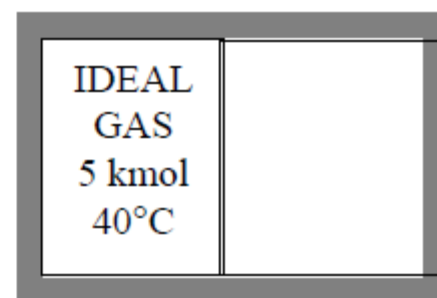
Analysis Taking the entire rigid tank as the system, the energy balance can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$0 = \Delta U = m(u_2 - u_1)$$

$$u_2 = u_1$$

$$T_2 = T_1$$



since $u = u(T)$ for an ideal gas. Then the entropy change of the gas becomes

$$\begin{aligned} \Delta S &= N \left(\bar{c}_{v,\text{avg}} \ln \frac{T_2}{T_1} + R_u \ln \frac{V_2}{V_1} \right) = NR_u \ln \frac{V_2}{V_1} \\ &= (5 \text{ kmol})(8.314 \text{ kJ/kmol} \cdot \text{K}) \ln(2) \\ &= \mathbf{28.81 \text{ kJ/K}} \end{aligned}$$

This also represents the **total entropy change** since the tank does not contain anything else, and there are no interactions with the surroundings.

7-75 Air is compressed in a piston–cylinder device from 100 kPa and 17°C to 800 kPa in a reversible, adiabatic process. Determine the final temperature and the work done during this process, assuming (a) constant specific heats and (b) variable specific heats for air. *Answers: (a) 525.3 K, 171.1 kJ/kg, (b) 522.4 K, 169.3 kJ/kg*

Analysis (a) Assuming constant specific heats, the ideal gas isentropic relations give

$$T_2 = T_1 \left(\frac{P_2}{P_1} \right)^{(k-1)/k} = (290 \text{ K}) \left(\frac{800 \text{ kPa}}{100 \text{ kPa}} \right)^{0.4/1.4} = \mathbf{525.3 \text{ K}}$$

Then,

$$T_{\text{avg}} = (290 + 525.3)/2 = 407.7 \text{ K} \longrightarrow c_{v,\text{avg}} = 0.727 \text{ kJ/kg} \cdot \text{K}$$

We take the air in the cylinder as the system. The energy balance for this stationary closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$W_{\text{in}} = \Delta U = m(u_2 - u_1) \cong mc_v(T_2 - T_1)$$

Thus,

$$w_{\text{in}} = c_{v,\text{avg}}(T_2 - T_1) = (0.727 \text{ kJ/kg} \cdot \text{K})(525.3 - 290) \text{ K} = \mathbf{171.1 \text{ kJ/kg}}$$

(b) Assuming variable specific heats, the final temperature can be determined using the relative pressure data (Table A-17),

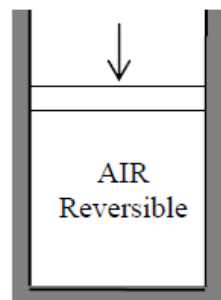
$$T_1 = 290 \text{ K} \longrightarrow \begin{matrix} P_{r_1} = 1.2311 \\ u_1 = 206.91 \text{ kJ/kg} \end{matrix}$$

and

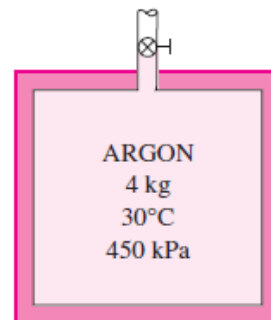
$$P_{r_2} = \frac{P_2}{P_1} P_{r_1} = \frac{800 \text{ kPa}}{100 \text{ kPa}} (1.2311) = 9.849 \longrightarrow \begin{matrix} T_2 = \mathbf{522.4 \text{ K}} \\ u_2 = 376.16 \text{ kJ/kg} \end{matrix}$$

Then the work input becomes

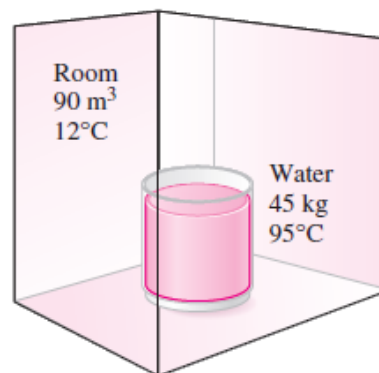
$$w_{\text{in}} = u_2 - u_1 = (376.16 - 206.91) \text{ kJ/kg} = \mathbf{169.25 \text{ kJ/kg}}$$



7-78 An insulated rigid tank contains 4 kg of argon gas at 450 kPa and 30°C. A valve is now opened, and argon is allowed to escape until the pressure inside drops to 200 kPa. Assuming the argon remaining inside the tank has undergone a reversible, adiabatic process, determine the final mass in the tank. *Answer: 2.46 kg*



7-83 A container filled with 45 kg of liquid water at 95°C is placed in a 90-m³ room that is initially at 12°C. Thermal equilibrium is established after a while as a result of heat transfer between the water and the air in the room. Using constant specific heats, determine (a) the final equilibrium temperature, (b) the amount of heat transfer between the water and the air in the room, and (c) the entropy generation. Assume the room is well sealed and heavily insulated.



7-84 Air at 800 kPa and 400°C enters a steady-flow nozzle with a low velocity and leaves at 100 kPa. If the air undergoes an adiabatic expansion process through the nozzle, what is the maximum velocity of the air at the nozzle exit, in m/s?

7-85 An ideal gas at 100 kPa and 27°C enters a steady-flow compressor. The gas is compressed to 400 kPa, and 10 percent of the mass that entered the compressor is removed for some other use. The remaining 90 percent of the inlet gas is compressed to 600 kPa before leaving the compressor. The entire compression process is assumed to be reversible and adiabatic. The power supplied to the compressor is measured to be 32 kW. If the ideal gas has constant specific heats such that $c_v = 0.8 \text{ kJ/kg} \cdot \text{K}$ and $c_p = 1.1 \text{ kJ/kg} \cdot \text{K}$, (a) sketch the compression process on a $T-s$ diagram, (b) determine the temperature of the gas at the two compressor exits, in K, and (c) determine the mass flow rate of the gas into the compressor, in kg/s.

7-86 A constant-volume tank contains 5 kg of air at 100 kPa and 327°C. The air is cooled to the surroundings temperature of 27°C. Assume constant specific heats at 300 K. (a) Determine the entropy change of the air in the tank during the process, in kJ/K, (b) determine the net entropy change of the universe due to this process, in kJ/K, and (c) sketch the processes for the air in the tank and the surroundings on a single T - s diagram. Be sure to label the initial and final states for both processes.

Assumptions 1 Air is an ideal gas with constant specific heats.

Properties The specific heat of air at room temperature is $c_v = 0.718 \text{ kJ/kg}\cdot\text{K}$ (Table A-2a).

Analysis (a) The entropy change of air is determined from

$$\begin{aligned}\Delta S_{\text{air}} &= mc_v \ln \frac{T_2}{T_1} \\ &= (5 \text{ kg})(0.718 \text{ kJ/kg}\cdot\text{K}) \ln \frac{(27 + 273) \text{ K}}{(327 + 273) \text{ K}} \\ &= \mathbf{-2.488 \text{ kJ/K}}\end{aligned}$$

(b) An energy balance on the system gives

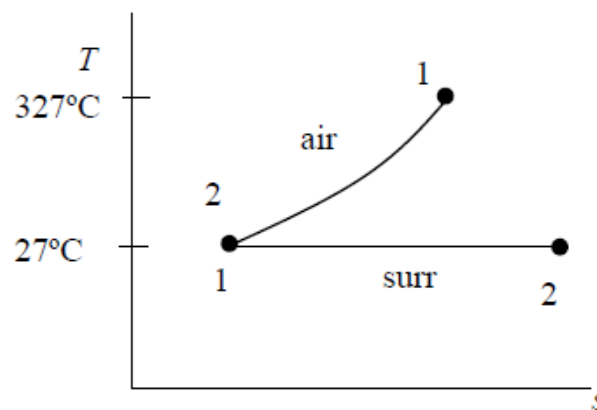
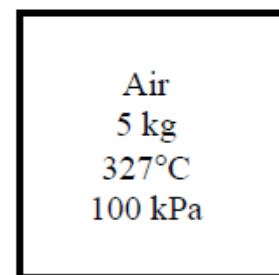
$$\begin{aligned}Q_{\text{out}} &= mc_v(T_2 - T_1) \\ &= (5 \text{ kg})(0.718 \text{ kJ/kg}\cdot\text{K})(327 - 27) \\ &= 1077 \text{ kJ}\end{aligned}$$

The entropy change of the surroundings is

$$\Delta S_{\text{surr}} = \frac{Q_{\text{out}}}{T_{\text{surr}}} = \frac{1077 \text{ kJ}}{300 \text{ K}} = 3.59 \text{ kJ/K}$$

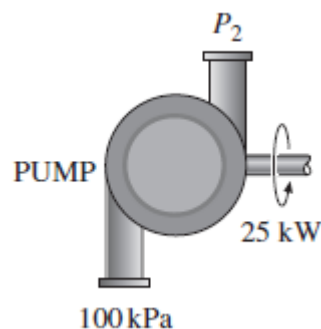
The entropy change of universe due to this process is

$$S_{\text{gen}} = \Delta S_{\text{total}} = \Delta S_{\text{air}} + \Delta S_{\text{surr}} = -2.488 + 3.59 = \mathbf{1.10 \text{ kJ/K}}$$



Reversible Steady-Flow Work

7-91 Liquid water enters a 25-kW pump at 100-kPa pressure at a rate of 5 kg/s. Determine the highest pressure the liquid water can have at the exit of the pump. Neglect the kinetic and potential energy changes of water, and take the specific volume of water to be 0.001 m³/kg. Answer: 5100 kPa



Analysis The highest pressure the liquid can have at the pump exit can be determined from the reversible steady-flow work relation for a liquid,

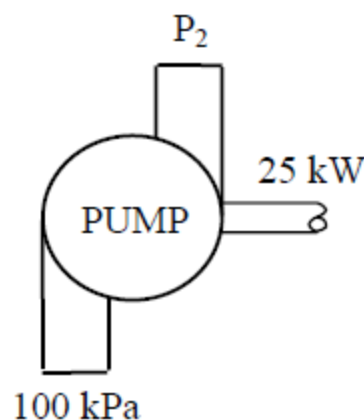
$$\dot{W}_{\text{in}} = \dot{m} \left(\int_1^2 v dP + \Delta ke^{\varphi_0} + \Delta pe^{\varphi_0} \right) = \dot{m} v_1 (P_2 - P_1)$$

Thus,

$$25 \text{ kJ/s} = (5 \text{ kg/s})(0.001 \text{ m}^3/\text{kg})(P_2 - 100) \text{ kPa} \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right)$$

It yields

$$P_2 = \mathbf{5100 \text{ kPa}}$$



7-93 Consider a steam power plant that operates between the pressure limits of 10 MPa and 20 kPa. Steam enters the pump as saturated liquid and leaves the turbine as saturated vapor. Determine the ratio of the work delivered by the turbine to the work consumed by the pump. Assume the entire cycle to be reversible and the heat losses from the pump and the turbine to be negligible.

Analysis Both the compression and expansion processes are reversible and adiabatic, and thus isentropic,

$s_1 = s_2$ and $s_3 = s_4$. Then the properties of the steam are

$$\left. \begin{array}{l} P_4 = 20 \text{ kPa} \\ \text{sat. vapor} \end{array} \right\} \begin{array}{l} h_4 = h_g @ 20 \text{ kPa} = 2608.9 \text{ kJ/kg} \\ s_4 = s_g @ 20 \text{ kPa} = 7.9073 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_3 = 10 \text{ MPa} \\ s_3 = s_4 \end{array} \right\} h_3 = 4707.2 \text{ kJ/kg}$$

Also, $\nu_1 = \nu_f @ 20 \text{ kPa} = 0.001017 \text{ m}^3/\text{kg}$.

The work output to this isentropic turbine is determined from the steady-flow energy balance to be

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{0 (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}h_3 = \dot{m}h_4 + \dot{W}_{\text{out}}$$

$$\dot{W}_{\text{out}} = \dot{m}(h_3 - h_4)$$

Substituting,

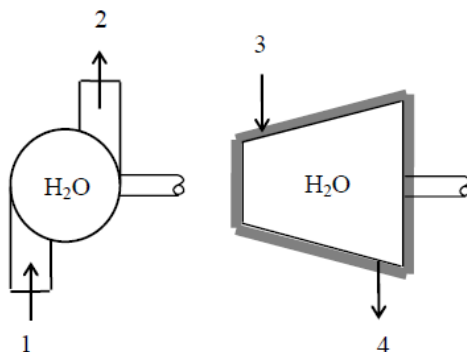
$$w_{\text{turb,out}} = h_3 - h_4 = 4707.2 - 2608.9 = 2098.3 \text{ kJ/kg}$$

The pump work input is determined from the steady-flow work relation to be

$$\begin{aligned} w_{\text{pump,in}} &= \int_1^2 \nu dP + \Delta ke^0 + \Delta pe^0 = \nu_1(P_2 - P_1) \\ &= (0.001017 \text{ m}^3/\text{kg})(10,000 - 20) \text{ kPa} \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) \\ &= 10.15 \text{ kJ/kg} \end{aligned}$$

Thus,

$$\frac{w_{\text{turb,out}}}{w_{\text{pump,in}}} = \frac{2098.3}{10.15} = \mathbf{206.7}$$



7-95 Liquid water at 120 kPa enters a 7-kW pump where its pressure is raised to 5 MPa. If the elevation difference between the exit and the inlet levels is 10 m, determine the highest mass flow rate of liquid water this pump can handle. Neglect the kinetic energy change of water, and take the specific volume of water to be $0.001 \text{ m}^3/\text{kg}$.

Isentropic Efficiencies of Steady-Flow Devices

7-106 Steam enters an adiabatic turbine at 7 MPa, 600°C, and 80 m/s and leaves at 50 kPa, 150°C, and 140 m/s. If the power output of the turbine is 6 MW, determine (a) the mass flow rate of the steam flowing through the turbine and (b) the isentropic efficiency of the turbine. *Answers: (a) 6.95 kg/s, (b) 73.4 percent*

Assumptions 1 This is a steady-flow process since there is no change with time. 2 Potential energy changes are negligible. 3 The device is adiabatic and thus heat transfer is negligible.

Analysis (a) From the steam tables (Tables A-4 and A-6),

$$\left. \begin{array}{l} P_1 = 7 \text{ MPa} \\ T_1 = 600^\circ\text{C} \end{array} \right\} \begin{array}{l} h_1 = 3650.6 \text{ kJ/kg} \\ s_1 = 7.0910 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_2 = 50 \text{ kPa} \\ T_2 = 150^\circ\text{C} \end{array} \right\} h_{2a} = 2780.2 \text{ kJ/kg}$$

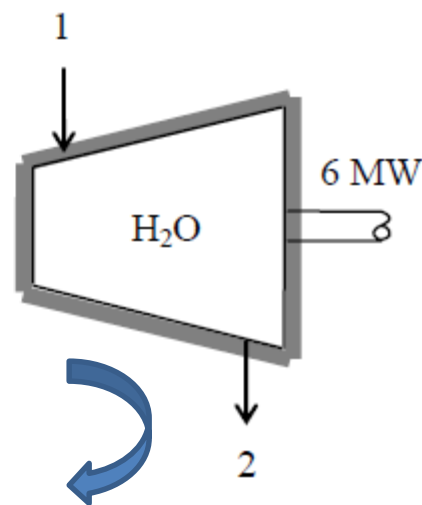
There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. We take the actual turbine as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\approx 0 \text{ (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}(h_1 + V_1^2/2) = \dot{W}_{\text{a,out}} + \dot{m}(h_2 + V_2^2/2) \quad (\text{since } \dot{Q} \cong \Delta \text{pe} \cong 0)$$

$$\dot{W}_{\text{a,out}} = -\dot{m} \left(h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} \right)$$



Substituting, the mass flow rate of the steam is determined to be

$$6000 \text{ kJ/s} = -\dot{m} \left(2780.2 - 3650.6 + \frac{(140 \text{ m/s})^2 - (80 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) \right)$$

$$\dot{m} = \mathbf{6.95 \text{ kg/s}}$$

(b) The isentropic exit enthalpy of the steam and the power output of the isentropic turbine are

$$\left. \begin{array}{l} P_{2s} = 50 \text{ kPa} \\ s_{2s} = s_1 \end{array} \right\} \quad x_{2s} = \frac{s_{2s} - s_f}{s_{fg}} = \frac{7.0910 - 1.0912}{6.5019} = 0.9228$$

$$h_{2s} = h_f + x_{2s} h_{fg} = 340.54 + (0.9228)(2304.7) = 2467.3 \text{ kJ/kg}$$

and

$$\dot{W}_{s,\text{out}} = -\dot{m} \left(h_{2s} - h_1 + \left\{ (V_2^2 - V_1^2) / 2 \right\} \right)$$

$$\dot{W}_{s,\text{out}} = -(6.95 \text{ kg/s}) \left(2467.3 - 3650.6 + \frac{(140 \text{ m/s})^2 - (80 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) \right)$$

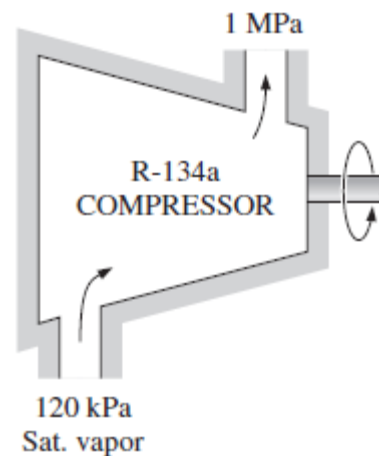
$$= 8174 \text{ kW}$$

Then the isentropic efficiency of the turbine becomes

$$\eta_r = \frac{\dot{W}_a}{\dot{W}_s} = \frac{6000 \text{ kW}}{8174 \text{ kW}} = 0.734 = \mathbf{73.4\%}$$

7-109

Refrigerant-134a enters an adiabatic compressor as saturated vapor at 120 kPa at a rate of $0.3 \text{ m}^3/\text{min}$ and exits at 1-MPa pressure. If the isentropic efficiency of the compressor is 80 percent, determine (a) the temperature of the refrigerant at the exit of the compressor and (b) the power input, in kW. Also, show the process on a T - s diagram with respect to saturation lines.



7-111 Air enters an adiabatic compressor at 100 kPa and 17°C at a rate of 2.4 m³/s, and it exits at 257°C. The compressor has an isentropic efficiency of 84 percent. Neglecting the changes in kinetic and potential energies, determine (a) the exit pressure of air and (b) the power required to drive the compressor.

Properties The gas constant of air is $R = 0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}$ (Table A-1)

Analysis (a) From the air table (Table A-17),

$$T_1 = 290 \text{ K} \longrightarrow h_1 = 290.16 \text{ kJ/kg}, \quad P_{r1} = 1.2311$$

$$T_2 = 530 \text{ K} \longrightarrow h_{2a} = 533.98 \text{ kJ/kg}$$

From the isentropic efficiency relation $\eta_c = \frac{h_{2s} - h_1}{h_{2a} - h_1}$,

$$\begin{aligned} h_{2s} &= h_1 + \eta_c (h_{2a} - h_1) \\ &= 290.16 + (0.84)(533.98 - 290.16) = 495.0 \text{ kJ/kg} \longrightarrow P_{r2} = 7.951 \end{aligned}$$

Then from the isentropic relation ,

$$\frac{P_2}{P_1} = \frac{P_{r2}}{P_{r1}} \longrightarrow P_2 = \left(\frac{P_{r2}}{P_{r1}} \right) P_1 = \left(\frac{7.951}{1.2311} \right) (100 \text{ kPa}) = \mathbf{646 \text{ kPa}}$$

(b) There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$. We take the actual compressor as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{Eq. (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{W}_{\text{a,in}} + \dot{m}h_1 = \dot{m}h_2 \quad (\text{since } \dot{Q} \cong \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

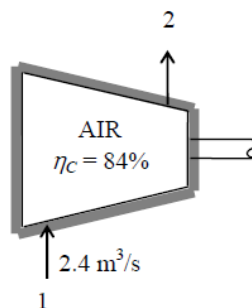
$$\dot{W}_{\text{a,in}} = \dot{m}(h_2 - h_1)$$

where

$$\dot{m} = \frac{P_1 \dot{V}_1}{RT_1} = \frac{(100 \text{ kPa})(2.4 \text{ m}^3/\text{s})}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(290 \text{ K})} = 2.884 \text{ kg/s}$$

Then the power input to the compressor is determined to be

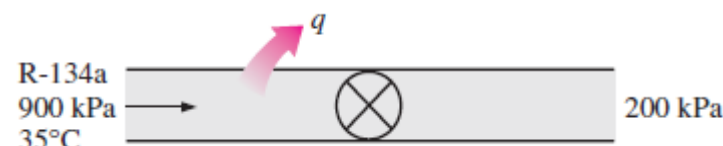
$$\dot{W}_{\text{a,in}} = (2.884 \text{ kg/s})(533.98 - 290.16) \text{ kJ/kg} = \mathbf{703 \text{ kW}}$$



7-112 Air is compressed by an adiabatic compressor from 95 kPa and 27°C to 600 kPa and 277°C. Assuming variable specific heats and neglecting the changes in kinetic and potential energies, determine (a) the isentropic efficiency of the compressor and (b) the exit temperature of air if the process were reversible. *Answers: (a) 81.9 percent, (b) 505.5 K*

Entropy Balance

7-118 Refrigerant-134a is throttled from 900 kPa and 35°C to 200 kPa. Heat is lost from the refrigerant in the amount of 0.8 kJ/kg to the surroundings at 25°C. Determine (a) the exit temperature of the refrigerant and (b) the entropy generation during this process.

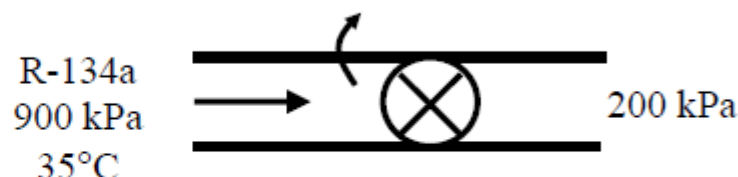


Analysis The properties of the refrigerant at the inlet of the device are (Table A-13)

$$\left. \begin{array}{l} P_1 = 900 \text{ kPa} \\ T_1 = 35^\circ\text{C} \end{array} \right\} \begin{array}{l} h_1 = 100.86 \text{ kJ/kg} \\ s_1 = 0.37135 \text{ kJ/kg} \end{array}$$

The enthalpy of the refrigerant at the exit of the device is

$$h_2 = h_1 - q_{\text{out}} = 100.86 - 0.8 = 100.06 \text{ kJ/kg}$$



Now, the properties at the exit state may be obtained from the R-134a tables

$$\left. \begin{array}{l} P_2 = 200 \text{ kPa} \\ h_2 = 100.06 \text{ kJ/kg} \end{array} \right\} \begin{array}{l} T_2 = -10.09^\circ\text{C} \\ s_2 = 0.38884 \text{ kJ/kg.K} \end{array}$$

The entropy generation associated with this process may be obtained by adding the entropy change of R-134a as it flows in the device and the entropy change of the surroundings.

$$\Delta s_{\text{R-134a}} = s_2 - s_1 = 0.38884 - 0.37135 = 0.01749 \text{ kJ/kg.K}$$

$$\Delta s_{\text{surr}} = \frac{q_{\text{out}}}{T_{\text{surr}}} = \frac{0.8 \text{ kJ/kg}}{(25 + 273) \text{ K}} = 0.002685 \text{ kJ/kg.K}$$

$$s_{\text{gen}} = \Delta s_{\text{total}} = \Delta s_{\text{R-134a}} + \Delta s_{\text{surr}} = 0.01749 + 0.002685 = \mathbf{0.0202 \text{ kJ/kg.K}}$$

7-121 A rigid tank contains 7.5 kg of saturated water mixture at 400 kPa. A valve at the bottom of the tank is now opened, and liquid is withdrawn from the tank. Heat is transferred to the steam such that the pressure inside the tank remains constant. The valve is closed when no liquid is left in the tank. If it is estimated that a total of 5 kJ of heat is transferred to the tank, determine (a) the quality of steam in the tank at the initial state, (b) the amount of mass that has escaped, and (c) the entropy generation during this process if heat is supplied to the tank from a source at 500°C.

Analysis (a) The properties of the steam in the tank at the final state and the properties of exiting steam are (Tables A-4 through A-6)

$$\left. \begin{array}{l} P_2 = 400 \text{ kPa} \\ x_2 = 1 \text{ (sat. vap.)} \end{array} \right\} \begin{array}{l} u_2 = 2553.1 \text{ kJ/kg} \\ v_2 = 0.46242 \text{ m}^3/\text{kg} \\ s_2 = 6.8955 \text{ kJ/kg}\cdot\text{K} \end{array}$$

$$\left. \begin{array}{l} P_{2e} = 400 \text{ kPa} \\ x_e = 0 \text{ (sat. liq.)} \end{array} \right\} \begin{array}{l} h_e = 604.66 \text{ kJ/kg} \\ s_e = 1.7765 \text{ kJ/kg}\cdot\text{K} \end{array}$$

The relations for the volume of the tank and the final mass in the tank are

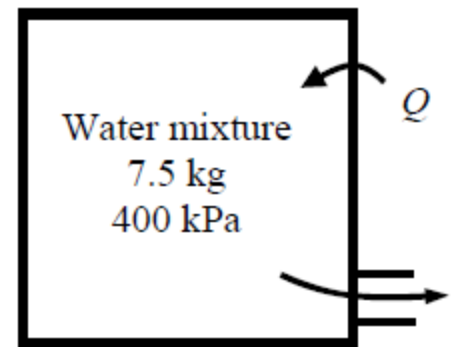
$$\begin{aligned} V &= m_1 v_1 = (7.5 \text{ kg})v_1 \\ m_2 &= \frac{V}{v_2} = \frac{(7.5 \text{ kg})v_1}{0.46242 \text{ m}^3/\text{kg}} = 16.219v_1 \end{aligned}$$

The mass, energy, and entropy balances may be written as

$$m_e = m_1 - m_2$$

$$Q_{\text{in}} - m_e h_e = m_2 u_2 - m_1 u_1$$

$$\frac{Q_{\text{in}}}{T_{\text{source}}} - m_e s_e + S_{\text{gen}} = m_2 s_2 - m_1 s_1$$



Substituting,

$$m_e = 7.5 - 16.219\nu_1 \quad (1)$$

$$5 - (7.5 - 16.219\nu_1)(604.66) = 16.219\nu_1(2553.1) - 7.5u_1 \quad (2)$$

$$\frac{5}{500 + 273} - (7.5 - 16.219\nu_1)(1.7765) + S_{\text{gen}} = 16.219\nu_1(6.8955) - 7.5s_1 \quad (3)$$

Eq. (2) may be solved by a trial-error approach by trying different qualities at the inlet state. Or, we can use EES to solve the equations to find

$$x_1 = \mathbf{0.8666}$$

Other properties at the initial state are

$$\left. \begin{array}{l} P_1 = 400 \text{ kPa} \\ x_1 = 0.8666 \end{array} \right\} \begin{array}{l} u_1 = 2293.2 \text{ kJ/kg} \\ \nu_1 = 0.40089 \text{ m}^3/\text{kg} \\ s_1 = 6.2129 \text{ kJ/kg.K} \end{array}$$

Substituting into Eqs (1) and (3),

$$(b) \quad m_e = 7.5 - 16.219(0.40089) = \mathbf{0.998 \text{ kg}}$$

(c)

$$\frac{5}{500 + 273} - [7.5 - 16.219(0.40089)](1.7765) + S_{\text{gen}} = 16.219(0.40089)(6.8955) - 7.5(6.2129)$$
$$S_{\text{gen}} = \mathbf{0.00553 \text{ kJ/K}}$$

7-122 Consider a family of four, with each person taking a 5-min shower every morning. The average flow rate through the shower head is 12 L/min. City water at 15°C is heated to 55°C in an electric water heater and tempered to 42°C by cold water at the T-elbow of the shower before being routed to the shower head. Determine the amount of entropy generated by this family per year as a result of taking daily showers.

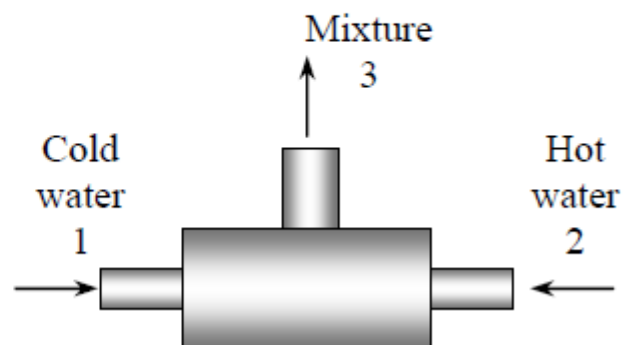
Properties The density and specific heat of water at room temperature are $\rho = 1 \text{ kg/L} = 1000 \text{ kg/m}^3$ and $c = 4.18 \text{ kJ/kg} \cdot ^\circ\text{C}$ (Table A-3).

Analysis The mass flow rate of water at the shower head is

$$\dot{m} = \rho \dot{V} = (1 \text{ kg/L})(12 \text{ L/min}) = 12 \text{ kg/min}$$

The mass balance for the mixing chamber can be expressed in the rate form as

$$\begin{aligned} \dot{m}_{\text{in}} - \dot{m}_{\text{out}} &= \Delta \dot{m}_{\text{system}} \stackrel{\text{no (steady)}}{=} 0 \\ \dot{m}_{\text{in}} &= \dot{m}_{\text{out}} \longrightarrow \dot{m}_1 + \dot{m}_2 = \dot{m}_3 \end{aligned}$$



where the subscript 1 denotes the cold water stream, 2 the hot water stream, and 3 the mixture.

The rate of entropy generation during this process can be determined by applying the rate form of the entropy balance on a system that includes the electric water heater and the mixing chamber (the T-elbow). Noting that there is no entropy transfer associated with work transfer (electricity) and there is no heat transfer, the entropy balance for this steady-flow system can be expressed as

$$\underbrace{\dot{S}_{\text{in}} - \dot{S}_{\text{out}}}_{\text{Rate of net entropy transfer by heat and mass}} + \underbrace{\dot{S}_{\text{gen}}}_{\text{Rate of entropy generation}} = \underbrace{\Delta \dot{S}_{\text{system}}}_{\text{Rate of change of entropy}} \stackrel{\text{no (steady)}}{=} 0$$



$$\dot{m}_1 s_1 + \dot{m}_2 s_2 - \dot{m}_3 s_3 + \dot{S}_{\text{gen}} = 0 \quad (\text{since } Q = 0 \text{ and work is entropy free})$$

$$\dot{S}_{\text{gen}} = \dot{m}_3 s_3 - \dot{m}_1 s_1 - \dot{m}_2 s_2$$

Noting from mass balance that $\dot{m}_1 + \dot{m}_2 = \dot{m}_3$ and $s_2 = s_1$ since hot water enters the system at the same temperature as the cold water, the rate of entropy generation is determined to be

$$\begin{aligned} \dot{S}_{\text{gen}} &= \dot{m}_3 s_3 - (\dot{m}_1 + \dot{m}_2) s_1 = \dot{m}_3 (s_3 - s_1) = \dot{m}_3 c_p \ln \frac{T_3}{T_1} \\ &= (12 \text{ kg/min})(4.18 \text{ kJ/kg.K}) \ln \frac{42 + 273}{15 + 273} = 4.495 \text{ kJ/min.K} \end{aligned}$$

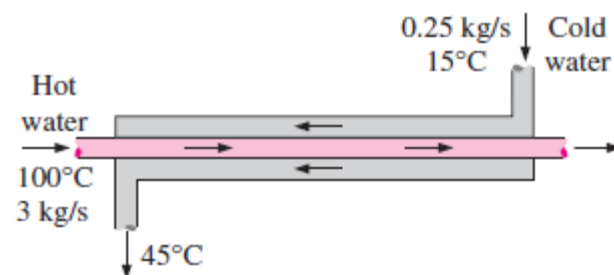
Noting that 4 people take a 5-min shower every day, the amount of entropy generated per year is

$$\begin{aligned} S_{\text{gen}} &= (\dot{S}_{\text{gen}}) \Delta t (\text{No. of people}) (\text{No. of days}) \\ &= (4.495 \text{ kJ/min.K})(5 \text{ min/person} \cdot \text{day})(4 \text{ persons})(365 \text{ days/year}) \\ &= \mathbf{32,814 \text{ kJ/K}} \quad (\text{per year}) \end{aligned}$$

Discussion The value above represents the entropy generated within the water heater and the T-elbow in the absence of any heat losses. It does not include the entropy generated as the shower water at 42°C is discarded or cooled to the outdoor temperature. Also, an entropy balance on the mixing chamber alone (hot water entering at 55°C instead of 15°C) will exclude the entropy generated within the water heater.

7-123 Steam is to be condensed in the condenser of a steam power plant at a temperature of 60°C with cooling water from a nearby lake, which enters the tubes of the condenser at 18°C at a rate of 75 kg/s and leaves at 27°C . Assuming the condenser to be perfectly insulated, determine (a) the rate of condensation of the steam and (b) the rate of entropy generation in the condenser. *Answers: (a) 1.20 kg/s , (b) 1.06 kW/K*

7-127 Cold water ($c_p = 4.18\text{ kJ/kg} \cdot ^{\circ}\text{C}$) leading to a shower enters a well-insulated, thin-walled, double-pipe, counter-flow heat exchanger at 15°C at a rate of 0.25 kg/s and is heated to 45°C by hot water ($c_p = 4.19\text{ kJ/kg} \cdot ^{\circ}\text{C}$) that enters at 100°C at a rate of 3 kg/s . Determine (a) the rate of heat transfer and (b) the rate of entropy generation in the heat exchanger.



7-128 Air ($c_p = 1.005 \text{ kJ/kg} \cdot ^\circ\text{C}$) is to be preheated by hot exhaust gases in a cross-flow heat exchanger before it enters the furnace. Air enters the heat exchanger at 95 kPa and 20°C at a rate of $1.6 \text{ m}^3/\text{s}$. The combustion gases ($c_p = 1.10 \text{ kJ/kg} \cdot ^\circ\text{C}$) enter at 180°C at a rate of 2.2 kg/s and leave at 95°C . Determine (a) the rate of heat transfer to the air, (b) the outlet temperature of the air, and (c) the rate of entropy generation.

Properties The specific heats of air and combustion gases are given to be 1.005 and $1.10 \text{ kJ/kg} \cdot ^\circ\text{C}$, respectively. The gas constant of air is $R = 0.287 \text{ kJ/kg} \cdot \text{K}$ (Table A-1).

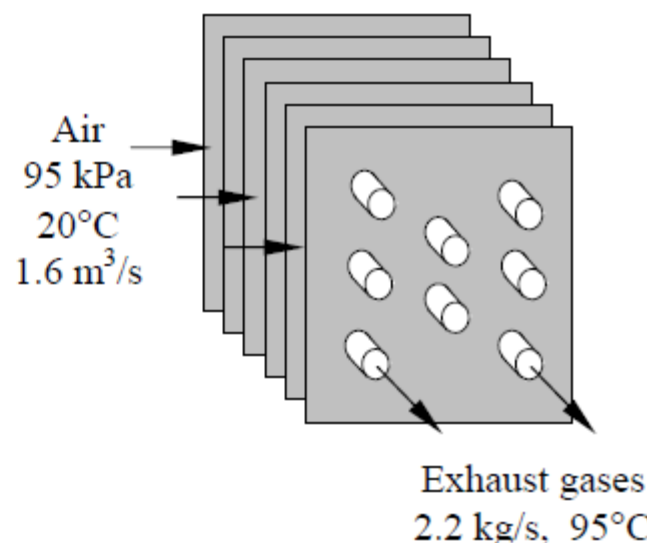
Analysis We take the exhaust pipes as the system, which is a control volume. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\dot{\Delta E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}h_1 = \dot{Q}_{\text{out}} + \dot{m}h_2 \quad (\text{since } \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

$$\dot{Q}_{\text{out}} = \dot{m}c_p(T_1 - T_2)$$



Then the rate of heat transfer from the exhaust gases becomes

$$\dot{Q} = [\dot{m}c_p(T_{\text{in}} - T_{\text{out}})]_{\text{gas}} = (2.2 \text{ kg/s})(1.1 \text{ kJ/kg} \cdot ^\circ\text{C})(180^\circ\text{C} - 95^\circ\text{C}) = \mathbf{205.7 \text{ kW}}$$

The mass flow rate of air is

$$\dot{m} = \frac{P\dot{V}}{RT} = \frac{(95 \text{ kPa})(1.6 \text{ m}^3/\text{s})}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K}) \times (293 \text{ K})} = 1.808 \text{ kg/s}$$



Noting that heat loss by the exhaust gases is equal to the heat gain by the air, the outlet temperature of the air becomes

$$\dot{Q} = \left[\dot{m} c_p (T_{\text{out}} - T_{\text{in}}) \right]_{\text{air}} \longrightarrow T_{\text{out}} = T_{\text{in}} + \frac{\dot{Q}}{\dot{m} c_p} = 20^\circ\text{C} + \frac{205.7 \text{ kW}}{(1.808 \text{ kg/s})(1.005 \text{ kJ/kg}\cdot^\circ\text{C})} = \mathbf{133.2^\circ\text{C}}$$

The rate of entropy generation within the heat exchanger is determined by applying the rate form of the entropy balance on the entire heat exchanger:

$$\underbrace{\dot{S}_{\text{in}} - \dot{S}_{\text{out}}}_{\text{Rate of net entropy transfer by heat and mass}} + \underbrace{\dot{S}_{\text{gen}}}_{\text{Rate of entropy generation}} = \underbrace{\Delta \dot{S}_{\text{system}}}_{\text{Rate of change of entropy}} \overset{\text{0 (steady)}}{=} 0$$

$$\dot{m}_1 s_1 + \dot{m}_3 s_3 - \dot{m}_2 s_2 - \dot{m}_4 s_4 + \dot{S}_{\text{gen}} = 0 \quad (\text{since } Q = 0)$$

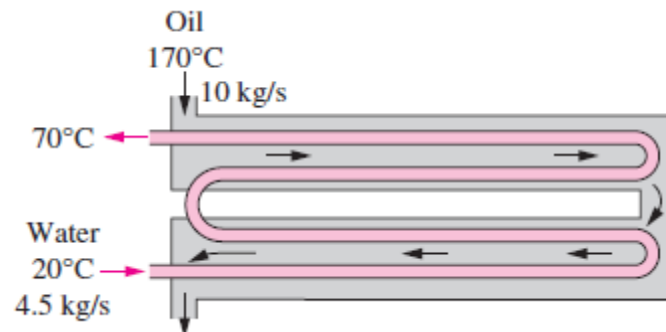
$$\dot{m}_{\text{exhaust}} s_1 + \dot{m}_{\text{air}} s_3 - \dot{m}_{\text{exhaust}} s_2 - \dot{m}_{\text{air}} s_4 + \dot{S}_{\text{gen}} = 0$$

$$\dot{S}_{\text{gen}} = \dot{m}_{\text{exhaust}} (s_2 - s_1) + \dot{m}_{\text{air}} (s_4 - s_3)$$

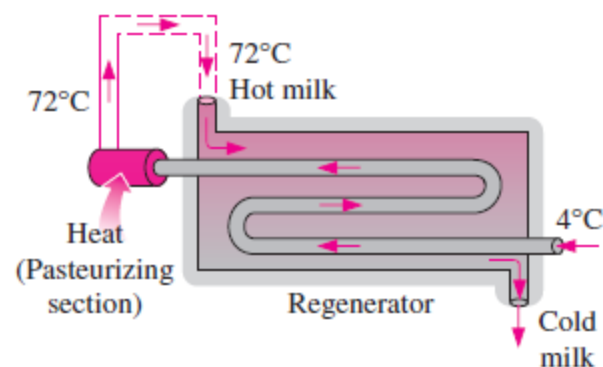
Then the rate of entropy generation is determined to be

$$\begin{aligned} \dot{S}_{\text{gen}} &= \dot{m}_{\text{exhaust}} c_p \ln \frac{T_2}{T_1} + \dot{m}_{\text{air}} c_p \ln \frac{T_4}{T_3} \\ &= (2.2 \text{ kg/s})(1.1 \text{ kJ/kg}\cdot\text{K}) \ln \frac{95 + 273}{180 + 273} + (1.808 \text{ kg/s})(1.005 \text{ kJ/kg}\cdot\text{K}) \ln \frac{133.2 + 273}{20 + 273} \\ &= \mathbf{0.091 \text{ kW/K}} \end{aligned}$$

7-129 A well-insulated, shell-and-tube heat exchanger is used to heat water ($c_p = 4.18 \text{ kJ/kg} \cdot ^\circ\text{C}$) in the tubes from 20 to 70°C at a rate of 4.5 kg/s . Heat is supplied by hot oil ($c_p = 2.30 \text{ kJ/kg} \cdot ^\circ\text{C}$) that enters the shell side at 170°C at a rate of 10 kg/s . Disregarding any heat loss from the heat exchanger, determine (a) the exit temperature of the oil and (b) the rate of entropy generation in the heat exchanger.



7-132 In a dairy plant, milk at 4°C is pasteurized continuously at 72°C at a rate of 12 L/s for 24 hours a day and 365 days a year. The milk is heated to the pasteurizing temperature by hot water heated in a natural-gas-fired boiler that has an efficiency of 82 percent. The pasteurized milk is then cooled by cold water at 18°C before it is finally refrigerated back to 4°C . To save energy and money, the plant installs a regenerator that has an effectiveness of 82 percent. If the cost of natural gas is $\$1.04/\text{therm}$ (1 therm = $105,500 \text{ kJ}$), determine how much energy and money the regenerator will save this company per year and the annual reduction in entropy generation.



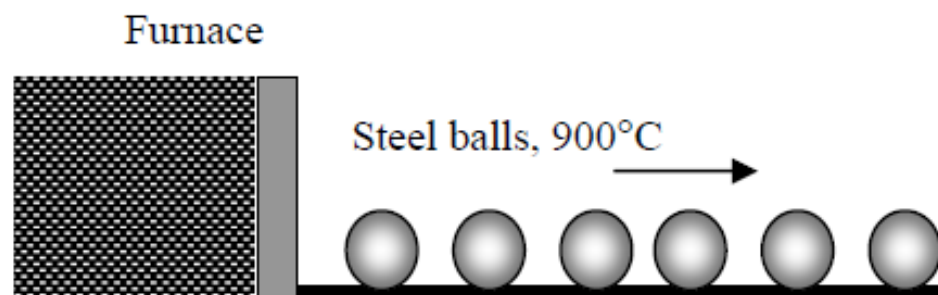
7-133 Stainless-steel ball bearings ($\rho = 8085 \text{ kg/m}^3$ and $c_p = 0.480 \text{ kJ/kg} \cdot ^\circ\text{C}$) having a diameter of 1.2 cm are to be quenched in water at a rate of 1400 per minute. The balls leave the oven at a uniform temperature of 900°C and are exposed to air at 30°C for a while before they are dropped into the water. If the temperature of the balls drops to 850°C prior to quenching, determine (a) the rate of heat transfer from the balls to the air and (b) the rate of entropy generation due to heat loss from the balls to the air.

Analysis (a) We take a single bearing ball as the system. The energy balance for this closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$-Q_{\text{out}} = \Delta U_{\text{ball}} = m(u_2 - u_1)$$

$$Q_{\text{out}} = mc(T_1 - T_2)$$



The total amount of heat transfer from a ball is

$$m = \rho V = \rho \frac{\pi D^3}{6} = (8085 \text{ kg/m}^3) \frac{\pi (0.012 \text{ m})^3}{6} = 0.007315 \text{ kg}$$

$$Q_{\text{out}} = mc(T_1 - T_2) = (0.007315 \text{ kg})(0.480 \text{ kJ/kg} \cdot ^\circ\text{C})(900 - 850)^\circ\text{C} = 0.1756 \text{ kJ/ball}$$

Then the rate of heat transfer from the balls to the air becomes

$$\dot{Q}_{\text{total}} = \dot{n}_{\text{ball}} Q_{\text{out (per ball)}} = (1400 \text{ balls/min}) \times (0.1756 \text{ kJ/ball}) = \mathbf{245.8 \text{ kJ/min} = 4.10 \text{ kW}}$$

Therefore, heat is lost to the air at a rate of 4.10 kW.



(b) We again take a single bearing ball as the system. The entropy generated during this process can be determined by applying an entropy balance on an *extended system* that includes the ball and its immediate surroundings so that the boundary temperature of the extended system is at 30°C at all times:

$$\underbrace{S_{\text{in}} - S_{\text{out}}}_{\text{Net entropy transfer by heat and mass}} + \underbrace{S_{\text{gen}}}_{\text{Entropy generation}} = \underbrace{\Delta S_{\text{system}}}_{\text{Change in entropy}}$$

$$-\frac{Q_{\text{out}}}{T_b} + S_{\text{gen}} = \Delta S_{\text{system}} \rightarrow S_{\text{gen}} = \frac{Q_{\text{out}}}{T_b} + \Delta S_{\text{system}}$$

where

$$\Delta S_{\text{system}} = m(s_2 - s_1) = mc_{\text{avg}} \ln \frac{T_2}{T_1} = (0.007315 \text{ kg})(0.480 \text{ kJ/kg}\cdot\text{K}) \ln \frac{850 + 273}{900 + 273} = -0.0001530 \text{ kJ/K}$$

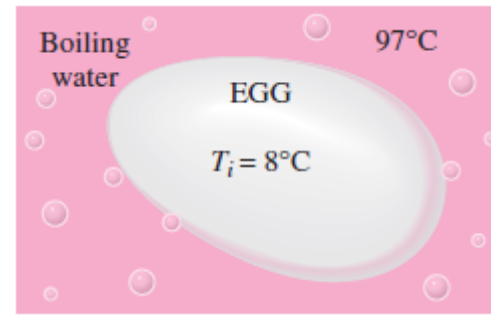
Substituting,

$$S_{\text{gen}} = \frac{Q_{\text{out}}}{T_b} + \Delta S_{\text{system}} = \frac{0.1756 \text{ kJ}}{303 \text{ K}} - 0.0001530 \text{ kJ/K} = 0.0004265 \text{ kJ/K (per ball)}$$

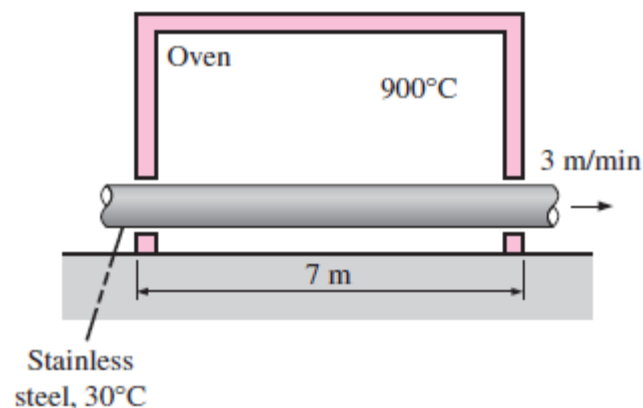
Then the rate of entropy generation becomes

$$\dot{S}_{\text{gen}} = S_{\text{gen}} \dot{n}_{\text{ball}} = (0.0004265 \text{ kJ/K} \cdot \text{ball})(1400 \text{ balls/min}) = 0.597 \text{ kJ/min}\cdot\text{K} = \mathbf{0.00995 \text{ kW/K}}$$

7-135 An ordinary egg can be approximated as a 5.5-cm-diameter sphere. The egg is initially at a uniform temperature of 8°C and is dropped into boiling water at 97°C . Taking the properties of the egg to be $\rho = 1020 \text{ kg/m}^3$ and $c_p = 3.32 \text{ kJ/kg} \cdot ^{\circ}\text{C}$, determine (a) how much heat is transferred to the egg by the time the average temperature of the egg rises to 70°C and (b) the amount of entropy generation associated with this heat transfer process.



7-137 Long cylindrical steel rods ($\rho = 7833 \text{ kg/m}^3$ and $c_p = 0.465 \text{ kJ/kg} \cdot ^\circ\text{C}$) of 10-cm diameter are heat treated by drawing them at a velocity of 3 m/min through a 7-m-long oven maintained at 900°C . If the rods enter the oven at 30°C and leave at 700°C , determine (a) the rate of heat transfer to the rods in the oven and (b) the rate of entropy generation associated with this heat transfer process.



Analysis (a) Noting that the rods enter the oven at a velocity of 3 m/min and exit at the same velocity, we can say that a 3-m long section of the rod is heated in the oven in 1 min. Then the mass of the rod heated in 1 minute is

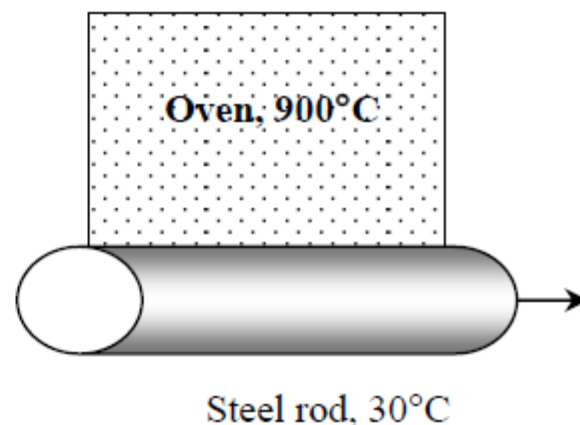
$$m = \rho V = \rho LA = \rho L(\pi D^2 / 4)$$

$$= (7833 \text{ kg/m}^3)(3 \text{ m})[\pi(0.1 \text{ m})^2 / 4] = 184.6 \text{ kg}$$

We take the 3-m section of the rod in the oven as the system. The energy balance for this closed system can be expressed as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$Q_{\text{in}} = \Delta U_{\text{rod}} = m(u_2 - u_1) = mc(T_2 - T_1)$$



Substituting,

$$Q_{\text{in}} = mc(T_2 - T_1) = (184.6 \text{ kg})(0.465 \text{ kJ/kg} \cdot ^\circ\text{C})(700 - 30)^\circ\text{C} = 57,512 \text{ kJ}$$

Noting that this much heat is transferred in 1 min, the rate of heat transfer to the rod becomes

$$\dot{Q}_{\text{in}} = Q_{\text{in}} / \Delta t = (57,512 \text{ kJ}) / (1 \text{ min}) = 57,512 \text{ kJ/min} = \mathbf{958.5 \text{ kW}}$$



(b) We again take the 3-m long section of the rod as the system. The entropy generated during this process can be determined by applying an entropy balance on an *extended system* that includes the rod and its immediate surroundings so that the boundary temperature of the extended system is at 900°C at all times:

$$\underbrace{\frac{S_{\text{in}} - S_{\text{out}}}{\text{Net entropy transfer by heat and mass}}}_{\text{Net entropy transfer by heat and mass}} + \underbrace{\frac{S_{\text{gen}}}{\text{Entropy generation}}}_{\text{Entropy generation}} = \underbrace{\Delta S_{\text{system}}}_{\text{Change in entropy}}$$

$$\frac{Q_{\text{in}}}{T_b} + S_{\text{gen}} = \Delta S_{\text{system}} \rightarrow S_{\text{gen}} = -\frac{Q_{\text{in}}}{T_b} + \Delta S_{\text{system}}$$

where

$$\Delta S_{\text{system}} = m(s_2 - s_1) = mc_{\text{avg}} \ln \frac{T_2}{T_1} = (184.6 \text{ kg})(0.465 \text{ kJ/kg.K}) \ln \frac{700 + 273}{30 + 273} = 100.1 \text{ kJ/K}$$

Substituting,

$$S_{\text{gen}} = -\frac{Q_{\text{in}}}{T_b} + \Delta S_{\text{system}} = -\frac{57,512 \text{ kJ}}{900 + 273 \text{ K}} + 100.1 \text{ kJ/K} = 51.1 \text{ kJ/K}$$

Noting that this much entropy is generated in 1 min, the rate of entropy generation becomes

$$\dot{S}_{\text{gen}} = \frac{S_{\text{gen}}}{\Delta t} = \frac{51.1 \text{ kJ/K}}{1 \text{ min}} = 51.1 \text{ kJ/min.K} = \mathbf{0.85 \text{ kW/K}}$$

7-138 The inner and outer surfaces of a $5\text{-m} \times 7\text{-m}$ brick wall of thickness 20 cm are maintained at temperatures of 20°C and 5°C , respectively. If the rate of heat transfer through the wall is 1515 W, determine the rate of entropy generation within the wall.

7-139 For heat transfer purposes, a standing man can be modeled as a 30-cm-diameter, 170-cm-long vertical cylinder with both the top and bottom surfaces insulated and with the side surface at an average temperature of 34°C . If the rate of heat loss from this man to the environment at 20°C is 336 W, determine the rate of entropy transfer from the body of this person accompanying heat transfer, in W/K.

7-140 A 1000-W iron is left on the ironing board with its base exposed to the air at 20°C . If the surface temperature is 400°C , determine the rate of entropy generation during this process in steady operation. How much of this entropy generation occurs within the iron?

7-143 Steam expands in a turbine steadily at a rate of 25,000 kg/h, entering at 6 MPa and 450°C and leaving at 20 kPa as saturated vapor. If the power generated by the turbine is 4 MW, determine the rate of entropy generation for this process. Assume the surrounding medium is at 25°C.

Answer: 11.0 kW/K

Properties From the steam tables (Tables A-4 through 6)

$$\begin{aligned} P_1 = 6 \text{ MPa} \quad \left. \begin{aligned} h_1 &= 3302.9 \text{ kJ/kg} \\ T_1 = 450^\circ\text{C} \quad s_1 &= 6.7219 \text{ kJ/kg} \cdot \text{K} \end{aligned} \right\} \\ P_2 = 20 \text{ kPa} \quad \left. \begin{aligned} h_2 &= 2608.9 \text{ kJ/kg} \\ \text{sat. vapor} \quad s_2 &= 7.9073 \text{ kJ/kg} \cdot \text{K} \end{aligned} \right\} \end{aligned}$$

Analysis There is only one inlet and one exit, and thus $\dot{m}_1 = \dot{m}_2 = \dot{m}$.

We take the turbine as the system, which is a control volume since mass crosses the boundary. The energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \overset{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}h_1 = \dot{Q}_{\text{out}} + \dot{W}_{\text{out}} + \dot{m}h_2$$

$$\dot{Q}_{\text{out}} = \dot{m}(h_1 - h_2) - \dot{W}_{\text{out}}$$

Substituting,

$$\dot{Q}_{\text{out}} = (25,000/3600 \text{ kg/s})(3302.9 - 2608.9) \text{ kJ/kg} - 4000 \text{ kJ/s} = 819.3 \text{ kJ/s}$$

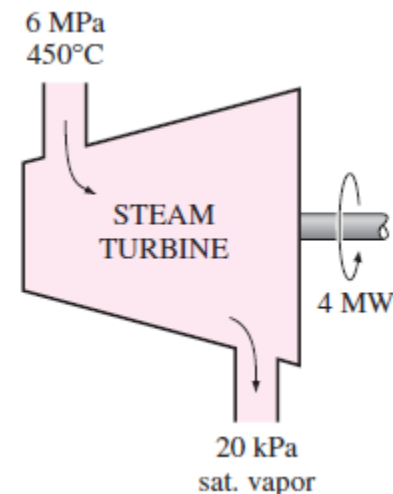
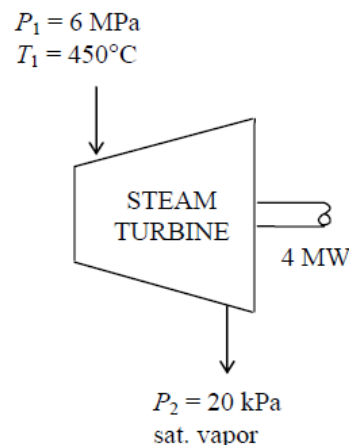
The rate of total entropy generation during this process is determined by applying the entropy balance on an *extended system* that includes the turbine and its immediate surroundings so that the boundary temperature of the extended system is 25°C at all times. It gives

$$\underbrace{\dot{S}_{\text{in}} - \dot{S}_{\text{out}}}_{\text{Rate of net entropy transfer by heat and mass}} + \underbrace{\dot{S}_{\text{gen}}}_{\text{Rate of entropy generation}} = \underbrace{\Delta \dot{S}_{\text{system}}}_{\text{Rate of change of entropy}} \overset{\text{no}}{=} 0$$

$$\dot{m}s_1 - \dot{m}s_2 - \frac{\dot{Q}_{\text{out}}}{T_{\text{b,surr}}} + \dot{S}_{\text{gen}} = 0$$

Substituting, the rate of entropy generation during this process is determined to be

$$\dot{S}_{\text{gen}} = \dot{m}(s_2 - s_1) + \frac{\dot{Q}_{\text{out}}}{T_{\text{b,surr}}} = (25,000/3600 \text{ kg/s})(7.9073 - 6.7219) \text{ kJ/kg} \cdot \text{K} + \frac{819.3 \text{ kW}}{298 \text{ K}} = \mathbf{11.0 \text{ kW/K}}$$



7-144 A hot-water stream at 70°C enters an adiabatic mixing chamber with a mass flow rate of 3.6 kg/s , where it is mixed with a stream of cold water at 20°C . If the mixture leaves the chamber at 42°C , determine (a) the mass flow rate of the cold water and (b) the rate of entropy generation during this adiabatic mixing process. Assume all the streams are at a pressure of 200 kPa .

7-145 Liquid water at 200 kPa and 20°C is heated in a chamber by mixing it with superheated steam at 200 kPa and 150°C. Liquid water enters the mixing chamber at a rate of 2.5 kg/s, and the chamber is estimated to lose heat to the surrounding air at 25°C at a rate of 1200 kJ/min. If the mixture leaves the mixing chamber at 200 kPa and 60°C, determine (a) the mass flow rate of the superheated steam and (b) the rate of entropy generation during this mixing process.

Answers: (a) 0.166 kg/s, (b) 0.333 kW/K

Properties Noting that $T < T_{\text{sat @ 200 kPa}} = 120.21^\circ\text{C}$, the cold water and the exit mixture streams exist as a compressed liquid, which can be approximated as a saturated liquid at the given temperature. From Tables A-4 through A-6,

$$\left. \begin{array}{l} P_1 = 200 \text{ kPa} \\ T_1 = 20^\circ\text{C} \end{array} \right\} \begin{array}{l} h_1 \cong h_{f@20^\circ\text{C}} = 83.91 \text{ kJ/kg} \\ s_1 \cong s_{f@20^\circ\text{C}} = 0.2965 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_2 = 200 \text{ kPa} \\ T_2 = 150^\circ\text{C} \end{array} \right\} \begin{array}{l} h_2 = 2769.1 \text{ kJ/kg} \\ s_2 = 7.2810 \text{ kJ/kg} \cdot \text{K} \end{array}$$

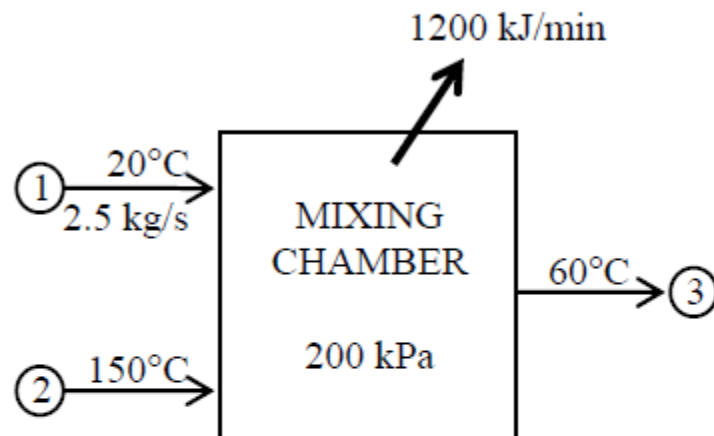
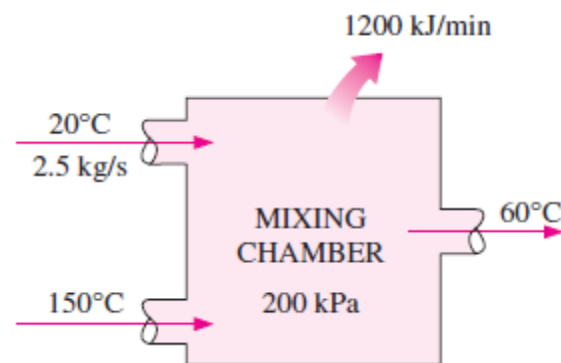
$$\left. \begin{array}{l} P_3 = 200 \text{ kPa} \\ T_3 = 60^\circ\text{C} \end{array} \right\} \begin{array}{l} h_3 \cong h_{f@60^\circ\text{C}} = 251.18 \text{ kJ/kg} \\ s_3 \cong s_{f@60^\circ\text{C}} = 0.8313 \text{ kJ/kg} \cdot \text{K} \end{array}$$

Analysis (a) We take the mixing chamber as the system, which is a control volume. The mass and energy balances for this steady-flow system can be expressed in the rate form as

$$\text{Mass balance: } \dot{m}_{\text{in}} - \dot{m}_{\text{out}} = \Delta \dot{E}_{\text{system}}^{\text{no (steady)}} = 0 \longrightarrow \dot{m}_1 + \dot{m}_2 = \dot{m}_3$$

Energy balance:

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}^{\text{no (steady)}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} = 0$$



$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}_1 h_1 + \dot{m}_2 h_2 = \dot{Q}_{\text{out}} + \dot{m}_3 h_3$$

Combining the two relations gives $\dot{Q}_{\text{out}} = \dot{m}_1 h_1 + \dot{m}_2 h_2 - (\dot{m}_1 + \dot{m}_2) h_3 = \dot{m}_1 (h_1 - h_3) + \dot{m}_2 (h_2 - h_3)$

Solving for $\dot{m}_2$ and substituting, the mass flow rate of the superheated steam is determined to be

$$\dot{m}_2 = \frac{\dot{Q}_{\text{out}} - \dot{m}_1 (h_1 - h_3)}{h_2 - h_3} = \frac{(1200/60 \text{ kJ/s}) - (2.5 \text{ kg/s})(83.91 - 251.18) \text{ kJ/kg}}{(2769.1 - 251.18) \text{ kJ/kg}} = \mathbf{0.166 \text{ kg/s}}$$

Also, $\dot{m}_3 = \dot{m}_1 + \dot{m}_2 = 2.5 + 0.166 = 2.666 \text{ kg/s}$

(b) The rate of total entropy generation during this process is determined by applying the entropy balance on an *extended system* that includes the mixing chamber and its immediate surroundings so that the boundary temperature of the extended system is 25°C at all times. It gives

$$\underbrace{\dot{S}_{\text{in}} - \dot{S}_{\text{out}}}_{\text{Rate of net entropy transfer by heat and mass}} + \underbrace{\dot{S}_{\text{gen}}}_{\text{Rate of entropy generation}} = \underbrace{\Delta \dot{S}_{\text{system}}}_{\text{Rate of change of entropy}} \overset{\text{Eq. 0}}{=} 0$$

$$\dot{m}_1 s_1 + \dot{m}_2 s_2 - \dot{m}_3 s_3 - \frac{\dot{Q}_{\text{out}}}{T_{\text{b,surr}}} + \dot{S}_{\text{gen}} = 0$$

Substituting, the rate of entropy generation during this process is determined to be

$$\begin{aligned} \dot{S}_{\text{gen}} &= \dot{m}_3 s_3 - \dot{m}_2 s_2 - \dot{m}_1 s_1 + \frac{\dot{Q}_{\text{out}}}{T_{\text{b,surr}}} \\ &= (2.666 \text{ kg/s})(0.8313 \text{ kJ/kg} \cdot \text{K}) - (0.166 \text{ kg/s})(7.2810 \text{ kJ/kg} \cdot \text{K}) \\ &\quad - (2.5 \text{ kg/s})(0.2965 \text{ kJ/kg} \cdot \text{K}) + \frac{(1200/60 \text{ kJ/s})}{298 \text{ K}} \\ &= \mathbf{0.333 \text{ kW/K}} \end{aligned}$$

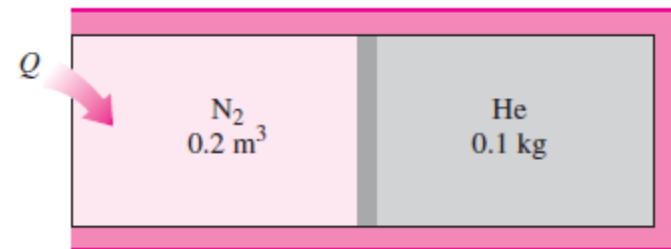
7-146 A 0.3-m^3 rigid tank is filled with saturated liquid water at 150°C . A valve at the bottom of the tank is now opened, and one-half of the total mass is withdrawn from the tank in the liquid form. Heat is transferred to water from a source at 200°C so that the temperature in the tank remains constant. Determine (a) the amount of heat transfer and (b) the total entropy generation for this process.

7-149 Steam enters an adiabatic nozzle at 4 MPa and 450°C with a velocity of 70 m/s and exits at 3 MPa and 320 m/s. If the nozzle has an inlet area of 7 cm^2 , determine (a) the exit temperature and (b) the rate of entropy generation for this process. *Answers: (a) 422.3°C , (b) 0.0361 kW/K*

Review Problems

7-162 A rigid tank contains 1.5 kg of water at 120°C and 500 kPa. Now 22 kJ of shaft work is done on the system and the final temperature in the tank is 95°C . If the entropy change of water is zero and the surroundings are at 15°C , determine (a) the final pressure in the tank, (b) the amount of heat transfer between the tank and the surroundings, and (c) the entropy generation during this process. *Answers: (a) 84.6 kPa, (b) 38.5 kJ, (c) 0.134 kJ/K*

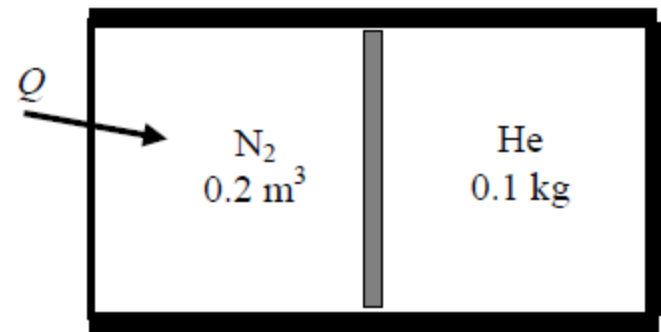
7-163 A horizontal cylinder is separated into two compartments by an adiabatic, frictionless piston. One side contains 0.2 m^3 of nitrogen and the other side contains 0.1 kg of helium, both initially at 20°C and 95 kPa . The sides of the cylinder and the helium end are insulated. Now heat is added to the nitrogen side from a reservoir at 500°C until the pressure of the helium rises to 120 kPa . Determine (a) the final temperature of the helium, (b) the final volume of the nitrogen, (c) the heat transferred to the nitrogen, and (d) the entropy generation during this process.



Properties The properties of nitrogen at room temperature are $R = 0.2968 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K}$, $c_p = 1.039 \text{ kJ/kg}\cdot\text{K}$, $c_v = 0.743 \text{ kJ/kg}\cdot\text{K}$, $k = 1.4$. The properties for helium are $R = 2.0769 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K}$, $c_p = 5.1926 \text{ kJ/kg}\cdot\text{K}$, $c_v = 3.1156 \text{ kJ/kg}\cdot\text{K}$, $k = 1.667$ (Table A-2).

Analysis (a) Helium undergoes an isentropic compression process, and thus the final helium temperature is determined from

$$T_{\text{He},2} = T_1 \left(\frac{P_2}{P_1} \right)^{(k-1)/k} = (20 + 273) \text{ K} \left(\frac{120 \text{ kPa}}{95 \text{ kPa}} \right)^{(1.667-1)/1.667} = \mathbf{321.7 \text{ K}}$$



(b) The initial and final volumes of the helium are

$$V_{\text{He},1} = \frac{mRT_1}{P_1} = \frac{(0.1 \text{ kg})(2.0769 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(20 + 273 \text{ K})}{95 \text{ kPa}} = 0.6406 \text{ m}^3$$

$$V_{\text{He},2} = \frac{mRT_2}{P_2} = \frac{(0.1 \text{ kg})(2.0769 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(321.7 \text{ K})}{120 \text{ kPa}} = 0.5568 \text{ m}^3$$

Then, the final volume of nitrogen becomes

$$V_{\text{N}_2,2} = V_{\text{N}_2,1} + V_{\text{He},1} - V_{\text{He},2} = 0.2 + 0.6406 - 0.5568 = \mathbf{0.2838 \text{ m}^3}$$



(c) The mass and final temperature of nitrogen are

$$m_{\text{N}_2} = \frac{P_1 V_1}{RT_1} = \frac{(95 \text{ kPa})(0.2 \text{ m}^3)}{(0.2968 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(20 + 273 \text{ K})} = 0.2185 \text{ kg}$$

$$T_{\text{N}_2,2} = \frac{P_2 V_2}{mR} = \frac{(120 \text{ kPa})(0.2838 \text{ m}^3)}{(0.2185 \text{ kg})(0.2968 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})} = 525.1 \text{ K}$$

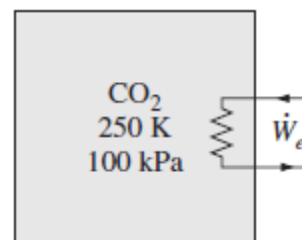
The heat transferred to the nitrogen is determined from an energy balance

$$\begin{aligned} Q_{\text{in}} &= \Delta U_{\text{N}_2} + \Delta U_{\text{He}} \\ &= [mc_v(T_2 - T_1)]_{\text{N}_2} + [mc_v(T_2 - T_1)]_{\text{He}} \\ &= (0.2185 \text{ kg})(0.743 \text{ kJ/kg} \cdot \text{K})(525.1 - 293) + (0.1 \text{ kg})(3.1156 \text{ kJ/kg} \cdot \text{K})(321.7 - 293) \\ &= \mathbf{46.6 \text{ kJ}} \end{aligned}$$

(d) Noting that helium undergoes an isentropic process, the entropy generation is determined to be

$$\begin{aligned} S_{\text{gen}} &= \Delta S_{\text{N}_2} + \Delta S_{\text{surr}} = m_{\text{N}_2} \left(c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \right) + \frac{-Q_{\text{in}}}{T_{\text{R}}} \\ &= (0.2185 \text{ kg}) \left[(1.039 \text{ kJ/kg} \cdot \text{K}) \ln \frac{525.1 \text{ K}}{293 \text{ K}} - (0.2968 \text{ kJ/kg} \cdot \text{K}) \ln \frac{120 \text{ kPa}}{95 \text{ kPa}} \right] + \frac{-46.6 \text{ kJ}}{(500 + 273) \text{ K}} \\ &= \mathbf{0.057 \text{ kJ/K}} \end{aligned}$$

7-164 A 0.8-m^3 rigid tank contains carbon dioxide (CO_2) gas at 250 K and 100 kPa . A 500-W electric resistance heater placed in the tank is now turned on and kept on for 40 min after which the pressure of CO_2 is measured to be 175 kPa . Assuming the surroundings to be at 300 K and using constant specific heats, determine (a) the final temperature of CO_2 , (b) the net amount of heat transfer from the tank, and (c) the entropy generation during this process.



7-165 Helium gas is throttled steadily from 500 kPa and 70°C . Heat is lost from the helium in the amount of 2.5 kJ/kg to the surroundings at 25°C and 100 kPa . If the entropy of the helium increases by $0.25\text{ kJ/kg} \cdot \text{K}$ in the valve, determine (a) the exit pressure and temperature and (b) the entropy generation during this process. *Answers: (a) 442 kPa , 69.5°C , (b) $0.258\text{ kJ/kg} \cdot \text{K}$*

7-166 Refrigerant-134a enters a compressor as a saturated vapor at 200 kPa at a rate of $0.03\text{ m}^3/\text{s}$ and leaves at 700 kPa . The power input to the compressor is 10 kW . If the surroundings at 20°C experience an entropy increase of 0.008 kW/K , determine (a) the rate of heat loss from the compressor, (b) the exit temperature of the refrigerant, and (c) the rate of entropy generation.

7-167 Air at 500 kPa and 400 K enters an adiabatic nozzle at a velocity of 30 m/s and leaves at 300 kPa and 350 K. Using variable specific heats, determine (a) the isentropic efficiency, (b) the exit velocity, and (c) the entropy generation.

Properties The gas constant of air is $R = 0.287 \text{ kJ/kg}\cdot\text{K}$ (Table A-1).

Assumptions 1 Steady operating conditions exist. 2 Potential energy changes are negligible.

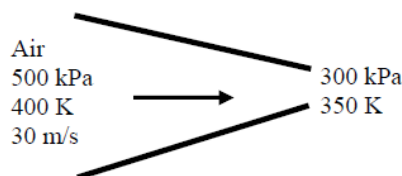
Analysis (a) (b) Using variable specific heats, the properties can be determined from air table as follows

$$T_1 = 400 \text{ K} \longrightarrow \begin{aligned} h_1 &= 400.98 \text{ kJ/kg} \\ s_1^0 &= 1.99194 \text{ kJ/kg}\cdot\text{K} \end{aligned}$$

$$P_{r1} = 3.806$$

$$T_2 = 350 \text{ K} \longrightarrow \begin{aligned} h_2 &= 350.49 \text{ kJ/kg} \\ s_2^0 &= 1.85708 \text{ kJ/kg}\cdot\text{K} \end{aligned}$$

$$P_{r2} = \frac{P_2}{P_1} P_{r1} = \frac{300 \text{ kPa}}{500 \text{ kPa}} (3.806) = 2.2836 \longrightarrow h_{2s} = 346.31 \text{ kJ/kg}$$



Energy balances on the control volume for the actual and isentropic processes give

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$$

$$400.98 \text{ kJ/kg} + \frac{(30 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) = 350.49 \text{ kJ/kg} + \frac{V_2^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right)$$

$$V_2 = \mathbf{319.1 \text{ m/s}}$$

$$h_1 + \frac{V_1^2}{2} = h_{2s} + \frac{V_{2s}^2}{2}$$

$$400.98 \text{ kJ/kg} + \frac{(30 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) = 346.31 \text{ kJ/kg} + \frac{V_{2s}^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right)$$

$$V_{2s} = 331.8 \text{ m/s}$$

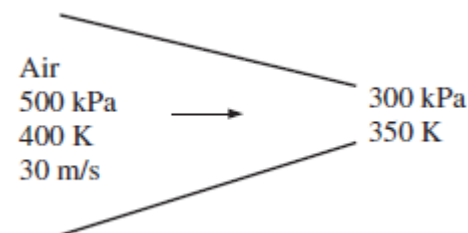
The isentropic efficiency is determined from its definition,

$$\eta_N = \frac{V_2^2}{V_{2s}^2} = \frac{(319.1 \text{ m/s})^2}{(331.8 \text{ m/s})^2} = \mathbf{0.925}$$

(c) Since the nozzle is adiabatic, the entropy generation is equal to the entropy increase of the air as it flows in the nozzle

$$s_{\text{gen}} = \Delta s_{\text{air}} = s_2^0 - s_1^0 - R \ln \frac{P_2}{P_1}$$

$$= (1.85708 - 1.99194) \text{ kJ/kg}\cdot\text{K} - (0.287 \text{ kJ/kg}\cdot\text{K}) \ln \frac{300 \text{ kPa}}{500 \text{ kPa}} = \mathbf{0.0118 \text{ kJ/kg}\cdot\text{K}}$$



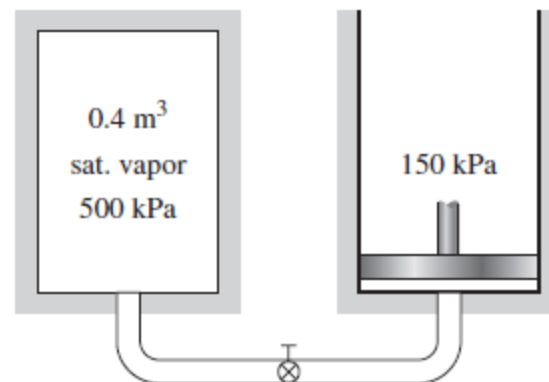
7-168 Show that the difference between the reversible steady-flow work and reversible moving boundary work is equal to the flow energy.

Analysis The total differential of flow energy $P\nu$ can be expressed as

$$d(P\nu) = P d\nu + \nu dP = \delta w_b - \delta w_{flow} = \delta(w_b - w_{flow})$$

Therefore, the difference between the reversible steady-flow work and the reversible boundary work is the flow energy.

7-169 An insulated tank containing 0.4 m^3 of saturated water vapor at 500 kPa is connected to an initially evacuated, insulated piston–cylinder device. The mass of the piston is such that a pressure of 150 kPa is required to raise it. Now the valve is opened slightly, and part of the steam flows to the cylinder, raising the piston. This process continues until the pressure in the tank drops to 150 kPa . Assuming the steam that remains in the tank to have undergone a reversible adiabatic process, determine the final temperature (a) in the rigid tank and (b) in the cylinder.



Analysis (a) The steam in tank A undergoes a reversible, adiabatic process, and thus $s_2 = s_1$. From the steam tables (Tables A-4 through A-6),

$$P_1 = 500 \text{ kPa} \left\{ \begin{array}{l} \nu_1 = \nu_{g@500 \text{ kPa}} = 0.37483 \text{ m}^3/\text{kg} \\ u_1 = u_{g@500 \text{ kPa}} = 2560.7 \text{ kJ/kg} \\ s_1 = s_{g@500 \text{ kPa}} = 6.8207 \text{ kJ/kg} \cdot \text{K} \end{array} \right. \text{ sat. vapor}$$

$$T_{2,A} = T_{\text{sat}@150 \text{ kPa}} = \mathbf{111.35^\circ\text{C}}$$

$$P_2 = 150 \text{ kPa} \left\{ \begin{array}{l} s_2 = s_1 \\ (\text{sat. mixture}) \end{array} \right. \quad \begin{array}{l} x_{2,A} = \frac{s_{2,A} - s_f}{s_{fg}} = \frac{6.8207 - 1.4337}{5.7894} = 0.9305 \\ \nu_{2,A} = \nu_f + x_{2,A} \nu_{fg} = 0.001053 + (0.9305)(1.1594 - 0.001053) = 1.0789 \text{ m}^3/\text{kg} \\ u_{2,A} = u_f + x_{2,A} u_{fg} = 466.97 + (0.9305)(2052.3 \text{ kJ/kg}) = 2376.6 \text{ kJ/kg} \end{array}$$

The initial and the final masses in tank A are

$$m_{1,A} = \frac{\nu_A}{\nu_{1,A}} = \frac{0.4 \text{ m}^3}{0.37483 \text{ m}^3/\text{kg}} = 1.067 \text{ kg} \quad \text{and} \quad m_{2,A} = \frac{\nu_A}{\nu_{2,A}} = \frac{0.4 \text{ m}^3}{1.0789 \text{ m}^3/\text{kg}} = 0.371 \text{ kg}$$

Thus,

$$m_{2,B} = m_{1,A} - m_{2,A} = 1.067 - 0.371 = 0.696 \text{ kg}$$

(b) The boundary work done during this process is

$$W_{b,out} = \int_1^2 P dV = P_B (v_{2,B} - 0) = P_B m_{2,B} v_{2,B}$$

Taking the contents of both the tank and the cylinder to be the system, the energy balance for this closed system can be expressed as

$$\underbrace{E_{in} - E_{out}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$-W_{b,out} = \Delta U = (\Delta U)_A + (\Delta U)_B$$

$$W_{b,out} + (\Delta U)_A + (\Delta U)_B = 0$$

$$\text{or, } P_B m_{2,B} v_{2,B} + (m_2 u_2 - m_1 u_1)_A + (m_2 u_2)_B = 0$$

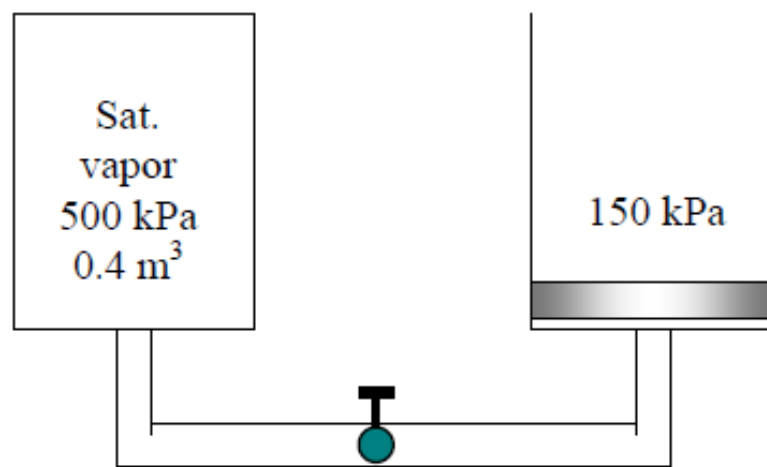
$$m_{2,B} h_{2,B} + (m_2 u_2 - m_1 u_1)_A = 0$$

Thus,

$$h_{2,B} = \frac{(m_1 u_1 - m_2 u_2)_A}{m_{2,B}} = \frac{(1.067)(2560.7) - (0.371)(2376.6)}{0.696} = 2658.8 \text{ kJ/kg}$$

At 150 kPa, $h_f = 467.13$ and $h_g = 2693.1$ kJ/kg. Thus at the final state, the cylinder will contain a saturated liquid-vapor mixture since $h_f < h_2 < h_g$. Therefore,

$$T_{2,B} = T_{\text{sat}@150 \text{ kPa}} = \mathbf{111.35^\circ \text{C}}$$



7-172 Air is compressed steadily by a compressor from 100 kPa and 17°C to 700 kPa at a rate of 5 kg/min. Determine the minimum power input required if the process is (a) adiabatic and (b) isothermal. Assume air to be an ideal gas with variable specific heats, and neglect the changes in kinetic and potential energies. *Answers: (a) 18.0 kW, (b) 13.5 kW*

Analysis (a) For the adiabatic case, the process will be reversible and adiabatic (i.e., isentropic), thus the isentropic relations are applicable.

$$T_1 = 290 \text{ K} \longrightarrow P_{r_1} = 1.2311 \text{ and } h_1 = 290.16 \text{ kJ/kg}$$

and

$$P_{r_2} = \frac{P_2}{P_1} P_{r_1} = \frac{700 \text{ kPa}}{100 \text{ kPa}} (1.2311) = 8.6177 \rightarrow \begin{matrix} T_2 = 503.3 \text{ K} \\ h_2 = 506.45 \text{ kJ/kg} \end{matrix}$$

The energy balance for the compressor, which is a steady-flow system, can be expressed in the rate form as

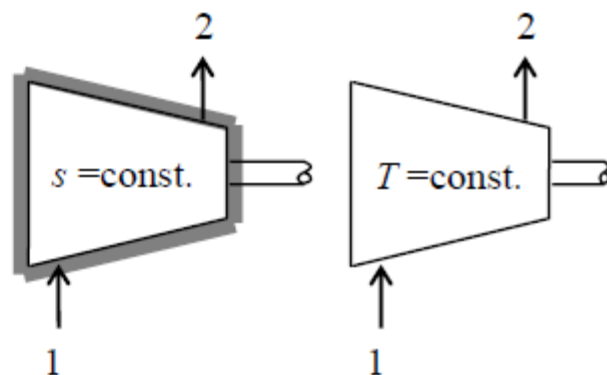
$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{W}_{\text{in}} + \dot{m}h_1 = \dot{m}h_2 \rightarrow \dot{W}_{\text{in}} = \dot{m}(h_2 - h_1)$$

Substituting, the power input to the compressor is determined to be

$$\dot{W}_{\text{in}} = (5/60 \text{ kg/s})(506.45 - 290.16) \text{ kJ/kg} = \mathbf{18.0 \text{ kW}}$$



(b) In the case of the reversible isothermal process, the steady-flow energy balance becomes

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}} \rightarrow \dot{W}_{\text{in}} + \dot{m}h_1 - \dot{Q}_{\text{out}} = \dot{m}h_2 \rightarrow \dot{W}_{\text{in}} = \dot{Q}_{\text{out}} + \dot{m}(h_2 - h_1)^{\text{g0}} = \dot{Q}_{\text{out}}$$

since $h = h(T)$ for ideal gases, and thus the enthalpy change in this case is zero. Also, for a reversible isothermal process,

$$\dot{Q}_{\text{out}} = \dot{m}T(s_1 - s_2) = -\dot{m}T(s_2 - s_1)$$

where

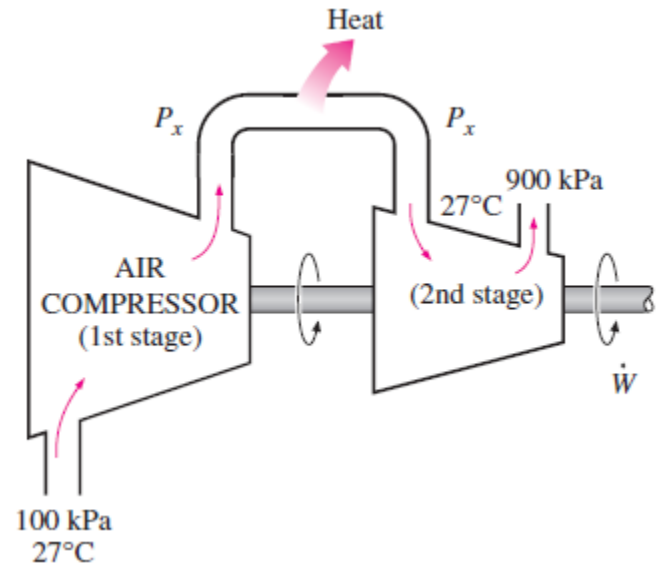
$$s_2 - s_1 = (s_2^{\circ} - s_1^{\circ})^{\text{g0}} - R \ln \frac{P_2}{P_1} = -R \ln \frac{P_2}{P_1} = -(0.287 \text{ kJ/kg} \cdot \text{K}) \ln \frac{700 \text{ kPa}}{100 \text{ kPa}} = -0.5585 \text{ kJ/kg} \cdot \text{K}$$

Substituting, the power input for the reversible isothermal case becomes

$$\dot{W}_{\text{in}} = -(5/60 \text{ kg/s})(290 \text{ K})(-0.5585 \text{ kJ/kg} \cdot \text{K}) = \mathbf{13.5 \text{ kW}}$$

7-173 Air enters a two-stage compressor at 100 kPa and 27°C and is compressed to 900 kPa. The pressure ratio across each stage is the same, and the air is cooled to the initial temperature between the two stages. Assuming the compression process to be isentropic, determine the power input to the compressor for a mass flow rate of 0.02 kg/s. What would your answer be if only one stage of compression were used?

Answers: 4.44 kW, 5.26 kW



7-174 Consider a three-stage isentropic compressor with two intercoolers that cool the gas to the initial temperature between the stages. Determine the two intermediate pressures (P_x and P_y) in terms of inlet and exit pressures (P_1 and P_2) that will minimize the work input to the compressor.

Answers: $P_x = (P_1^2 P_2)^{1/3}$, $P_y = (P_1 P_2^2)^{1/3}$

Analysis The work input to this three-stage compressor with intermediate pressures P_x and P_y and two intercoolers can be expressed as

$$\begin{aligned} w_{\text{comp}} &= w_{\text{comp,I}} + w_{\text{comp,II}} + w_{\text{comp,III}} \\ &= \frac{nRT_1}{n-1} \left(1 - (P_x/P_1)^{(n-1)/n} \right) + \frac{nRT_1}{n-1} \left(1 - (P_y/P_x)^{(n-1)/n} \right) + \frac{nRT_1}{n-1} \left(1 - (P_2/P_y)^{(n-1)/n} \right) \\ &= \frac{nRT_1}{n-1} \left(1 - (P_x/P_1)^{(n-1)/n} + 1 - (P_y/P_x)^{(n-1)/n} + 1 - (P_2/P_y)^{(n-1)/n} \right) \\ &= \frac{nRT_1}{n-1} \left(3 - (P_x/P_1)^{(n-1)/n} - (P_y/P_x)^{(n-1)/n} - (P_2/P_y)^{(n-1)/n} \right) \end{aligned}$$

The P_x and P_y values that will minimize the work input are obtained by taking the partial differential of w with respect to P_x and P_y , and setting them equal to zero:

$$\begin{aligned} \frac{\partial w}{\partial P_x} = 0 &\longrightarrow -\frac{n-1}{n} \left(\frac{1}{P_1} \right) \left(\frac{P_x}{P_1} \right)^{\frac{n-1}{n}-1} + \frac{n-1}{n} \left(\frac{1}{P_y} \right) \left(\frac{P_x}{P_y} \right)^{\frac{n-1}{n}-1} = 0 \\ \frac{\partial w}{\partial P_y} = 0 &\longrightarrow -\frac{n-1}{n} \left(\frac{1}{P_x} \right) \left(\frac{P_y}{P_x} \right)^{\frac{n-1}{n}-1} + \frac{n-1}{n} \left(\frac{1}{P_2} \right) \left(\frac{P_y}{P_2} \right)^{\frac{n-1}{n}-1} = 0 \end{aligned}$$

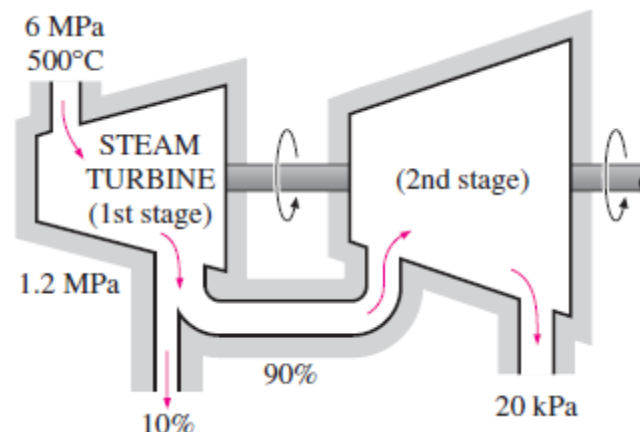
Simplifying,

$$\begin{aligned} \frac{1}{P_1} \left(\frac{P_x}{P_1} \right)^{\frac{1}{n}} &= \frac{1}{P_y} \left(\frac{P_x}{P_y} \right)^{\frac{2n-1}{n}} \longrightarrow \frac{1}{P_1^n} \left(\frac{P_x}{P_x} \right) = \frac{1}{P_y^n} \left(\frac{P_x}{P_y} \right)^{1-2n} \longrightarrow P_x^{2(1-n)} = (P_1 P_y)^{1-n} \\ \frac{1}{P_x} \left(\frac{P_y}{P_x} \right)^{\frac{1}{n}} &= \frac{1}{P_2} \left(\frac{P_y}{P_2} \right)^{\frac{2n-1}{n}} \longrightarrow \frac{1}{P_x^n} \left(\frac{P_y}{P_y} \right) = \frac{1}{P_2^n} \left(\frac{P_y}{P_2} \right)^{1-2n} \longrightarrow P_y^{2(1-n)} = (P_x P_2)^{1-n} \end{aligned}$$

which yield

$$\begin{aligned} P_x^2 &= P_1 \sqrt{P_x P_2} \longrightarrow P_x = (P_1^2 P_2)^{1/3} \\ P_y^2 &= P_2 \sqrt{P_1 P_y} \longrightarrow P_y = (P_1 P_2^2)^{1/3} \end{aligned}$$

7-175 Steam at 6 MPa and 500°C enters a two-stage adiabatic turbine at a rate of 15 kg/s. Ten percent of the steam is extracted at the end of the first stage at a pressure of 1.2 MPa for other use. The remainder of the steam is further expanded in the second stage and leaves the turbine at 20 kPa. Determine the power output of the turbine, assuming (a) the process is reversible and (b) the turbine has an isentropic efficiency of 88 percent. **Answers: (a) 16,291 kW, (b) 14,336 kW**



Properties From the steam tables (Tables A-4 through 6)

$$\left. \begin{array}{l} P_1 = 6 \text{ MPa} \\ T_1 = 500^\circ\text{C} \end{array} \right\} \begin{array}{l} h_1 = 3423.1 \text{ kJ/kg} \\ s_1 = 6.8826 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_2 = 1.2 \text{ MPa} \\ s_2 = s_1 \end{array} \right\} h_2 = 2962.8 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_3 = 20 \text{ kPa} \\ s_3 = s_1 \end{array} \right\} \begin{array}{l} x_{3s} = \frac{s_{3s} - s_f}{s_{fg}} = \frac{6.8826 - 0.8320}{7.0752} = 0.8552 \\ h_{3s} = h_f + x_{3s}h_{fg} = 251.42 + (0.8552)(2357.5) = 2267.5 \text{ kJ/kg} \end{array}$$

Analysis (a) The mass flow rate through the second stage is

$$\dot{m}_3 = 0.9\dot{m}_1 = (0.9)(15 \text{ kg/s}) = 13.5 \text{ kg/s}$$

We take the entire turbine, including the connection part between the two stages, as the system, which is a control volume since mass crosses the boundary. Noting that one fluid stream enters the turbine and two fluid streams leave, the energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\neq 0 \text{ (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}_1 h_1 = (\dot{m}_1 - \dot{m}_3) h_2 + \dot{W}_{\text{out}} + \dot{m}_3 h_3$$

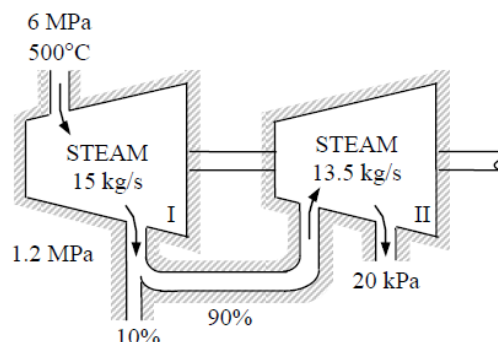
$$\begin{aligned} \dot{W}_{\text{out}} &= \dot{m}_1 h_1 - (\dot{m}_1 - \dot{m}_3) h_2 - \dot{m}_3 h_3 \\ &= \dot{m}_1 (h_1 - h_2) + \dot{m}_3 (h_2 - h_3) \end{aligned}$$

Substituting, the power output of the turbine is

$$\dot{W}_{\text{out}} = (15 \text{ kg/s})(3423.1 - 2962.8) \text{ kJ/kg} + (13.5 \text{ kg/s})(2962.8 - 2267.5) \text{ kJ/kg} = \mathbf{16,291 \text{ kW}}$$

(b) If the turbine has an adiabatic efficiency of 88%, then the power output becomes

$$\dot{W}_a = \eta_T \dot{W}_s = (0.88)(16,291 \text{ kW}) = \mathbf{14,336 \text{ kW}}$$

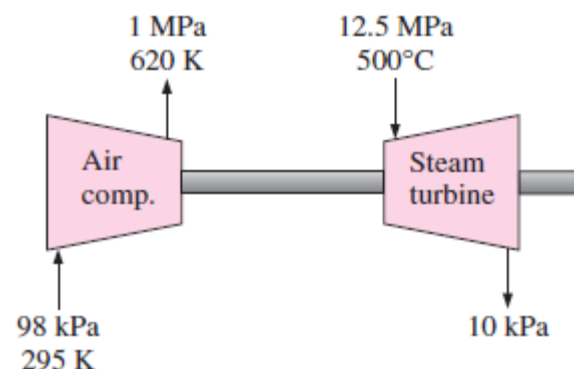


7-176 Steam enters a two-stage adiabatic turbine at 8 MPa and 550°C. It expands in the first stage to a pressure of 2 MPa. Then steam is reheated at constant pressure to 550°C before it is expanded in a second stage to a pressure of 200 kPa. The power output of the turbine is 80 MW. Assuming an isentropic efficiency of 84 percent for each stage of the turbine, determine the required mass flow rate of steam. Also, show the process on a T - s diagram with respect to saturation lines. *Answer: 85.8 kg/s*

7-177 Refrigerant-134a at 140 kPa and -10°C is compressed by an adiabatic 0.7-kW compressor to an exit state of 700 kPa and 50°C . Neglecting the changes in kinetic and potential energies, determine (a) the isentropic efficiency of the compressor, (b) the volume flow rate of the refrigerant at the compressor inlet, in L/min, and (c) the maximum volume flow rate at the inlet conditions that this adiabatic 0.7-kW compressor can handle without violating the second law.



An adiabatic air compressor is to be powered by a direct-coupled adiabatic steam turbine that is also driving a generator. Steam enters the turbine at 12.5 MPa and 500°C at a rate of 25 kg/s and exits at 10 kPa and a quality of 0.92. Air enters the compressor at 98 kPa and 295 K at a rate of 10 kg/s and exits at 1 MPa and 620 K. Determine (a) the net power delivered to the generator by the turbine and (b) the rate of entropy generation within the turbine and the compressor during this process.



Analysis There is only one inlet and one exit for either device, and thus $\dot{m}_{\text{in}} = \dot{m}_{\text{out}} = \dot{m}$. We take either the turbine or the compressor as the system, which is a control volume since mass crosses the boundary. The energy balance for either steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\approx 0 \text{ (steady)}}{=} 0 \rightarrow \dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

For the turbine and the compressor it becomes

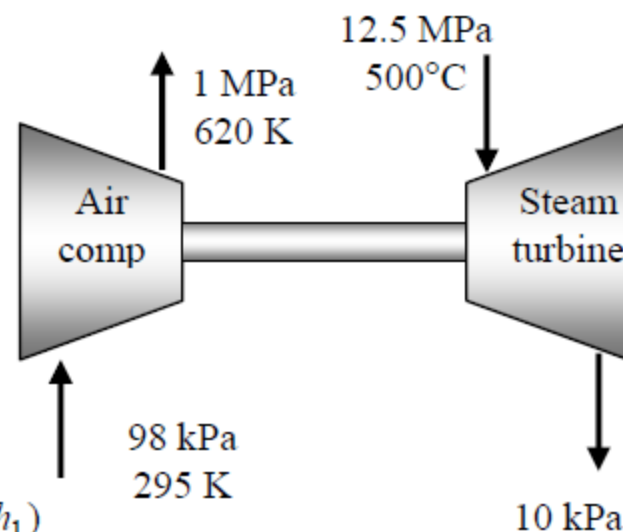
Compressor: $\dot{W}_{\text{comp,in}} + \dot{m}_{\text{air}} h_1 = \dot{m}_{\text{air}} h_2 \rightarrow \dot{W}_{\text{comp,in}} = \dot{m}_{\text{air}} (h_2 - h_1)$

Turbine: $\dot{m}_{\text{steam}} h_3 = \dot{W}_{\text{turb,out}} + \dot{m}_{\text{steam}} h_4 \rightarrow \dot{W}_{\text{turb,out}} = \dot{m}_{\text{steam}} (h_3 - h_4)$

Substituting,

$$\dot{W}_{\text{comp,in}} = (10 \text{ kg/s})(628.07 - 295.17) \text{ kJ/kg} = 3329 \text{ kW}$$

$$\dot{W}_{\text{turb,out}} = (25 \text{ kg/s})(3343.6 - 2392.5) \text{ kJ/kg} = 23,777 \text{ kW}$$



Therefore,

$$\dot{W}_{\text{net,out}} = \dot{W}_{\text{turb,out}} - \dot{W}_{\text{comp,in}} = 23,777 - 3329 = \mathbf{20,448 \text{ kW}}$$

Noting that the system is adiabatic, the total rate of entropy change (or generation) during this process is the sum of the entropy changes of both fluids,

$$\dot{S}_{\text{gen}} = \dot{m}_{\text{air}}(s_2 - s_1) + \dot{m}_{\text{steam}}(s_4 - s_3)$$

where

$$\begin{aligned} \dot{m}_{\text{air}}(s_2 - s_1) &= \dot{m} \left(s_2^\circ - s_1^\circ - R \ln \frac{P_2}{P_1} \right) \\ &= (10 \text{ kg/s}) \left(2.44356 - 1.68515 - 0.287 \ln \frac{1000 \text{ kPa}}{98 \text{ kPa}} \right) \text{kJ/kg} \cdot \text{K} = 0.92 \text{ kW/K} \end{aligned}$$

$$\dot{m}_{\text{steam}}(s_4 - s_3) = (25 \text{ kg/s})(7.5489 - 6.4651) \text{kJ/kg} \cdot \text{K} = 27.1 \text{ kW/K}$$

Substituting, the total rate of entropy generation is determined to be

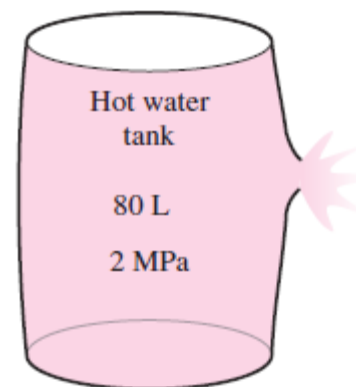
$$\dot{S}_{\text{gen,total}} = \dot{S}_{\text{gen,comp}} + \dot{S}_{\text{gen,turb}} = 0.92 + 27.1 = \mathbf{28.02 \text{ kW/K}}$$

7-182 The explosion of a hot water tank in a school in Spencer, Oklahoma, in 1982 killed 7 people while injuring 33 others. Although the number of such explosions has decreased dramatically since the development of the ASME Pressure Vessel Code, which requires the tanks to be designed to withstand four times the normal operating pressures, they still occur as a result of the failure of the pressure relief valves and thermostats. When a tank filled with a high-pressure and high-temperature liquid ruptures, the sudden drop of the pressure of the liquid to the atmospheric level causes part of the liquid to flash into vapor, and thus to experience a huge rise in its volume. The resulting pressure wave that propagates rapidly can cause considerable damage.

Considering that the pressurized liquid in the tank eventually reaches equilibrium with its surroundings shortly after the explosion, the work that a pressurized liquid would do if allowed to expand reversibly and adiabatically to the pressure of the surroundings can be viewed as the *explosive energy* of the pressurized liquid. Because of the very short time period of the explosion and the apparent calm afterward, the explosion process can be considered to be adiabatic with no changes in kinetic and potential energies and no mixing with the air.

Consider a 80-L hot-water tank that has a working pressure of 0.5 MPa. As a result of some malfunction, the pressure in the tank rises to 2 MPa, at which point the tank explodes. Taking the atmospheric pressure to be 100 kPa and assuming the liquid in the tank to be saturated at the time of explosion, determine the total explosion energy of the tank in terms of the TNT equivalence. (The explosion energy of TNT is about 3250 kJ/kg, and 5 kg of TNT can cause total destruction of unreinforced structures within about a 7-m radius.) *Answer:*

1.972 kg TNT



7-183 Using the arguments in the Prob. 7-182, determine the total explosion energy of a 0.35-L canned drink that explodes at a pressure of 1.2 MPa. To how many kg of TNT is this explosion energy equivalent?

7-184 Demonstrate the validity of the Clausius inequality using a reversible and an irreversible heat engine operating between the same two thermal energy reservoirs at constant temperatures of T_L and T_H .

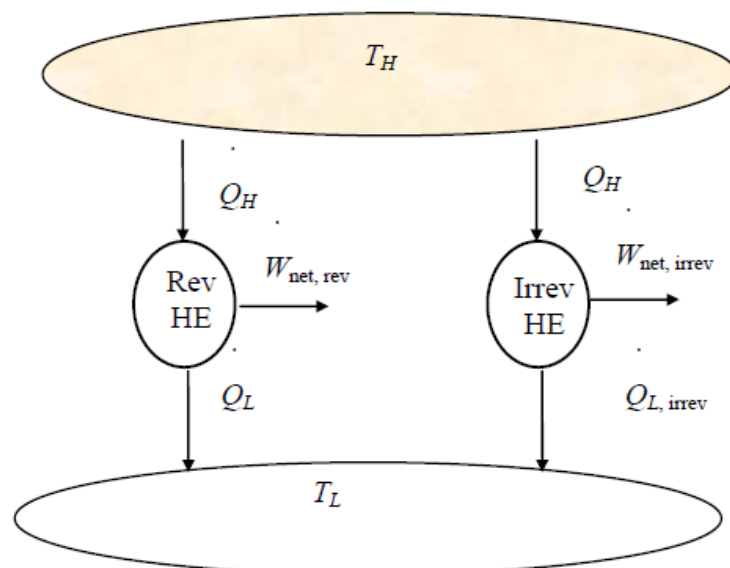
Analysis Consider two heat engines, one reversible and one irreversible, both operating between a high-temperature reservoir at T_H and a low-temperature reservoir at T_L . Both heat engines receive the same amount of heat, Q_H . The reversible heat engine rejects heat in the amount of Q_L , and the irreversible one in the amount of $Q_{L, \text{irrev}} = Q_L + Q_{\text{diff}}$, where Q_{diff} is a positive quantity since the irreversible heat engine produces less work. Noting that Q_H and Q_L are transferred at constant temperatures of T_H and T_L , respectively, the cyclic integral of $\delta Q/T$ for the reversible and irreversible heat engine cycles become

$$\oint \left(\frac{\delta Q}{T} \right)_{\text{rev}} = \int \frac{\delta Q_H}{T_H} - \int \frac{\delta Q_L}{T_L} = \frac{1}{T_H} \int \delta Q_H - \frac{1}{T_L} \int \delta Q_L = \frac{Q_H}{T_H} - \frac{Q_L}{T_L} = 0$$

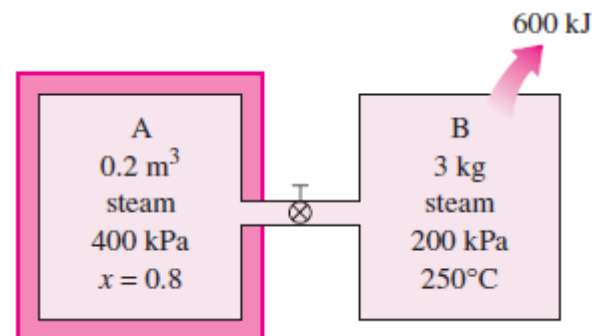
since $(Q_H/T_H) = (Q_L/T_L)$ for reversible cycles. Also,

$$\oint \left(\frac{\delta Q}{T} \right)_{\text{irrev}} = \frac{Q_H}{T_H} - \frac{Q_{L, \text{irrev}}}{T_L} = \frac{Q_H}{T_H} - \frac{Q_L}{T_L} - \frac{Q_{\text{diff}}}{T_L} = -\frac{Q_{\text{diff}}}{T_L} < 0$$

since Q_{diff} is a positive quantity. Thus, $\oint \left(\frac{\delta Q}{T} \right) \leq 0$.



7-186 Two rigid tanks are connected by a valve. Tank A is insulated and contains 0.2 m³ of steam at 400 kPa and 80 percent quality. Tank B is uninsulated and contains 3 kg of steam at 200 kPa and 250°C. The valve is now opened, and steam flows from tank A to tank B until the pressure in tank A drops to 300 kPa. During this process 600 kJ of heat is transferred from tank B to the surroundings at 0°C. Assuming the steam remaining inside tank A to have undergone a reversible adiabatic process, determine (a) the final temperature in each tank and (b) the entropy generated during this process. *Answers: (a) 133.5°C, 113.2°C; (b) 0.916 kJ/K*



Analysis (a) The steam in tank A undergoes a reversible, adiabatic process, and thus $s_2 = s_1$. From the steam tables (Tables A-4 through A-6),

Tank A:

$$P_1 = 400 \text{ kPa} \quad \left\{ \begin{array}{l} v_{1,A} = v_f + x_1 v_{fg} = 0.001084 + (0.8)(0.46242 - 0.001084) = 0.37015 \text{ m}^3/\text{kg} \\ u_{1,A} = u_f + x_1 u_{fg} = 604.22 + (0.8)(1948.9) = 2163.3 \text{ kJ/kg} \\ s_{1,A} = s_f + x_1 s_{fg} = 1.7765 + (0.8)(5.1191) = 5.8717 \text{ kJ/kg} \cdot \text{K} \end{array} \right.$$

$$T_{2,A} = T_{\text{sat}@300 \text{ kPa}} = \mathbf{133.52^\circ\text{C}}$$

$$P_1 = 300 \text{ kPa} \quad \left\{ \begin{array}{l} x_{2,A} = \frac{s_{2,A} - s_f}{s_{fg}} = \frac{5.8717 - 1.6717}{5.3200} = 0.7895 \\ v_{2,A} = v_f + x_{2,A} v_{fg} = 0.001073 + (0.7895)(0.60582 - 0.001073) = 0.47850 \text{ m}^3/\text{kg} \\ u_{2,A} = u_f + x_{2,A} u_{fg} = 561.11 + (0.7895)(1982.1 \text{ kJ/kg}) = 2125.9 \text{ kJ/kg} \end{array} \right.$$



Tank B:

$$\left. \begin{array}{l} P_1 = 200 \text{ kPa} \\ T_1 = 250^\circ\text{C} \end{array} \right\} \begin{array}{l} \nu_{1,B} = 1.1989 \text{ m}^3/\text{kg} \\ u_{1,B} = 2731.4 \text{ kJ/kg} \\ s_{1,B} = 7.7100 \text{ kJ/kg}\cdot\text{K} \end{array}$$

The initial and the final masses in tank A are

$$m_{1,A} = \frac{V_A}{\nu_{1,A}} = \frac{0.2 \text{ m}^3}{0.37015 \text{ m}^3/\text{kg}} = 0.5403 \text{ kg}$$

and

$$m_{2,A} = \frac{V_A}{\nu_{2,A}} = \frac{0.2 \text{ m}^3}{0.47850 \text{ m}^3/\text{kg}} = 0.4180 \text{ kg}$$

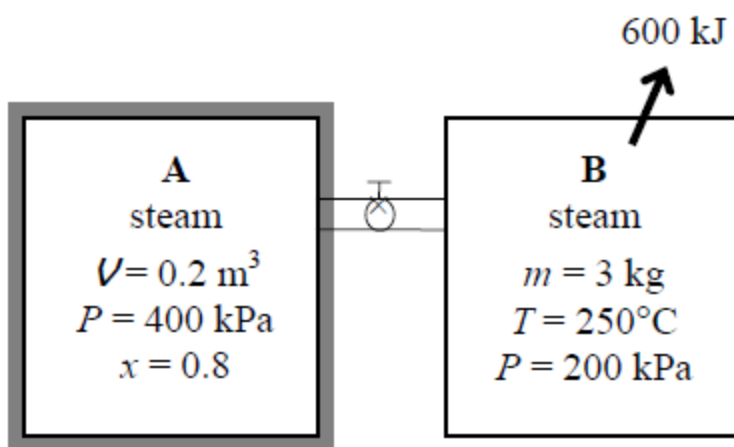
Thus, $0.5403 - 0.4180 = 0.1223 \text{ kg}$ of mass flows into tank B. Then,

$$m_{2,B} = m_{1,B} + 0.1223 = 3 + 0.1223 = 3.1223 \text{ kg}$$

The final specific volume of steam in tank B is determined from

$$\nu_{2,B} = \frac{V_B}{m_{2,B}} = \frac{(m_1 \nu_1)_B}{m_{2,B}} = \frac{(3 \text{ kg})(1.1989 \text{ m}^3/\text{kg})}{3.1223 \text{ kg}} = 1.1519 \text{ m}^3/\text{kg}$$

We take the entire contents of both tanks as the system, which is a closed system. The energy balance for this stationary closed system can be expressed as



$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$-Q_{\text{out}} = \Delta U = (\Delta U)_A + (\Delta U)_B \quad (\text{since } W = \text{KE} = \text{PE} = 0)$$

$$-Q_{\text{out}} = (m_2 u_2 - m_1 u_1)_A + (m_2 u_2 - m_1 u_1)_B$$

Substituting,

$$-600 = \{(0.418)(2125.9) - (0.5403)(2163.3)\} + \{(3.1223)u_{2,B} - (3)(2731.4)\}$$

$$u_{2,B} = 2522.0 \text{ kJ/kg}$$

Thus,

$$\left. \begin{array}{l} v_{2,B} = 1.1519 \text{ m}^3/\text{kg} \\ u_{2,B} = 2522.0 \text{ kJ/kg} \end{array} \right\} \begin{array}{l} T_{2,B} = \mathbf{113.2^\circ\text{C}} \\ s_{2,B} = 7.2274 \text{ kJ/kg} \cdot \text{K} \end{array}$$

(b) The total entropy generation during this process is determined by applying the entropy balance on an *extended system* that includes both tanks and their immediate surroundings so that the boundary temperature of the extended system is the temperature of the surroundings at all times. It gives

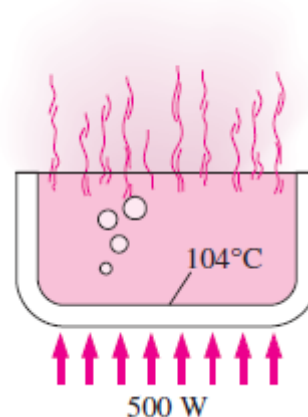
$$\underbrace{S_{\text{in}} - S_{\text{out}}}_{\text{Net entropy transfer by heat and mass}} + \underbrace{S_{\text{gen}}}_{\text{Entropy generation}} = \underbrace{\Delta S_{\text{system}}}_{\text{Change in entropy}}$$

$$-\frac{Q_{\text{out}}}{T_{\text{b,surr}}} + S_{\text{gen}} = \Delta S_A + \Delta S_B$$

Rearranging and substituting, the total entropy generated during this process is determined to be

$$\begin{aligned} S_{\text{gen}} &= \Delta S_A + \Delta S_B + \frac{Q_{\text{out}}}{T_{\text{b,surr}}} = (m_2 s_2 - m_1 s_1)_A + (m_2 s_2 - m_1 s_1)_B + \frac{Q_{\text{out}}}{T_{\text{b,surr}}} \\ &= \{(0.418)(5.8717) - (0.5403)(5.8717)\} + \{(3.1223)(7.2274) - (3)(7.7100)\} + \frac{600 \text{ kJ}}{273 \text{ K}} \\ &= \mathbf{0.916 \text{ kJ/K}} \end{aligned}$$

7-187 Heat is transferred steadily to boiling water in the pan through its flat bottom at a rate of 500 W. If the temperatures of the inner and outer surfaces of the bottom of the tank are 104°C and 105°C , respectively, determine the rate of entropy generation within bottom of the pan, in W/K.



7-188 A 1200-W electric resistance heating element whose diameter is 0.5 cm is immersed in 40 kg of water initially at 20°C . Assuming the water container is well-insulated, determine how long it will take for this heater to raise the water temperature to 50°C . Also, determine the entropy generated during this process, in kJ/K.

7-190 In large steam power plants, the feedwater is frequently heated in closed feedwater heaters, which are basically heat exchangers, by steam extracted from the turbine at some stage. Steam enters the feedwater heater at 1 MPa and 200°C and leaves as saturated liquid at the same pressure. Feedwater enters the heater at 2.5 MPa and 50°C and leaves 10°C below the exit temperature of the steam. Neglecting any heat losses from the outer surfaces of the heater, determine (a) the ratio of the mass flow rates of the extracted steam and the feedwater heater and (b) the total entropy change for this process per unit mass of the feedwater.

Properties The properties of steam and feedwater are (Tables A-4 through A-6)

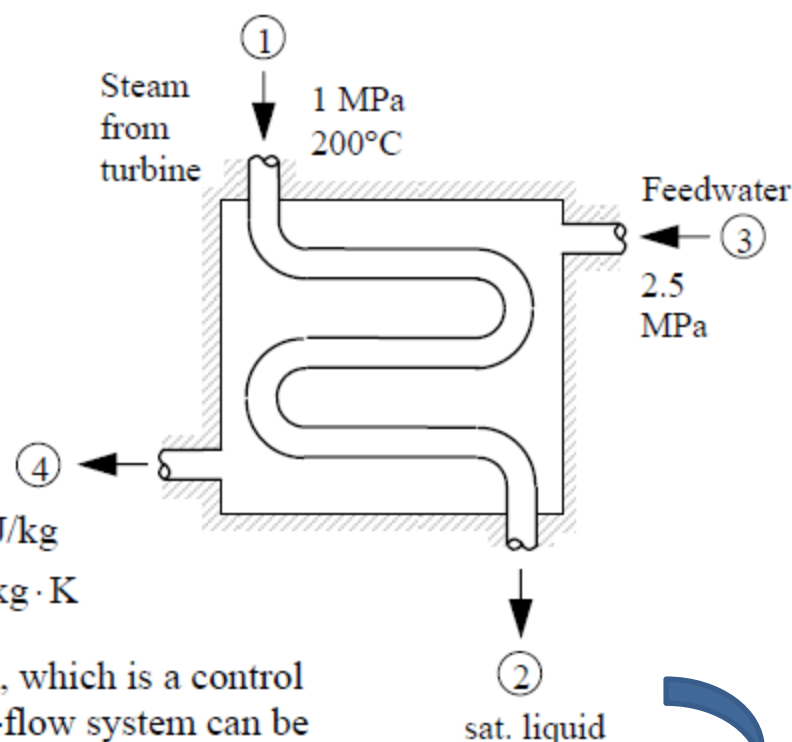
$$\left. \begin{array}{l} P_1 = 1 \text{ MPa} \\ T_1 = 200^\circ\text{C} \end{array} \right\} \begin{array}{l} h_1 = 2828.3 \text{ kJ/kg} \\ s_1 = 6.6956 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_2 = 1 \text{ MPa} \\ \text{sat. liquid} \end{array} \right\} \begin{array}{l} h_2 = h_f @ 1 \text{ MPa} = 762.51 \text{ kJ/kg} \\ s_2 = s_f @ 1 \text{ MPa} = 2.1381 \text{ kJ/kg} \cdot \text{K} \\ T_2 = 179.88^\circ\text{C} \end{array}$$

$$\left. \begin{array}{l} P_3 = 2.5 \text{ MPa} \\ T_3 = 50^\circ\text{C} \end{array} \right\} \begin{array}{l} h_3 \cong h_f @ 50^\circ\text{C} = 209.34 \text{ kJ/kg} \\ s_3 \cong s_f @ 50^\circ\text{C} = 0.7038 \text{ kJ/kg} \cdot \text{K} \end{array}$$

$$\left. \begin{array}{l} P_4 = 2.5 \text{ MPa} \\ T_4 = T_2 - 10^\circ\text{C} \cong 170^\circ\text{C} \end{array} \right\} \begin{array}{l} h_4 \cong h_f @ 170^\circ\text{C} = 719.08 \text{ kJ/kg} \\ s_4 \cong s_f @ 170^\circ\text{C} = 2.0417 \text{ kJ/kg} \cdot \text{K} \end{array}$$

Analysis (a) We take the heat exchanger as the system, which is a control volume. The mass and energy balances for this steady-flow system can be expressed in the rate form as follows:



Mass balance (for each fluid stream):

$$\dot{m}_{\text{in}} - \dot{m}_{\text{out}} = \Delta \dot{m}_{\text{system}} \stackrel{\text{no (steady)}}{=} 0 \rightarrow \dot{m}_{\text{in}} = \dot{m}_{\text{out}} \rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m}_s \quad \text{and} \quad \dot{m}_3 = \dot{m}_4 = \dot{m}_{fw}$$

Energy balance (for the heat exchanger):

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{Rate of change in internal, kinetic, potential, etc. energies}} \stackrel{\text{no (steady)}}{=} 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}_1 h_1 + \dot{m}_3 h_3 = \dot{m}_2 h_2 + \dot{m}_4 h_4 \quad (\text{since } \dot{Q} = \dot{W} = \Delta \text{ke} \cong \Delta \text{pe} \cong 0)$$

Combining the two, $\dot{m}_s (h_2 - h_1) = \dot{m}_{fw} (h_3 - h_4)$

Dividing by $\dot{m}_{fw}$ and substituting, $\frac{\dot{m}_s}{\dot{m}_{fw}} = \frac{h_4 - h_3}{h_1 - h_2} = \frac{(719.08 - 209.34) \text{ kJ/kg}}{(2828.3 - 762.51) \text{ kJ/kg}} = \mathbf{0.247}$

(b) The total entropy change (or entropy generation) during this process per unit mass of feedwater can be determined from an entropy balance expressed in the rate form as

$$\underbrace{\dot{S}_{\text{in}} - \dot{S}_{\text{out}}}_{\text{Rate of net entropy transfer by heat and mass}} + \underbrace{\dot{S}_{\text{gen}}}_{\text{Rate of entropy generation}} = \underbrace{\Delta \dot{S}_{\text{system}}}_{\text{Rate of change of entropy}} \stackrel{\text{no}}{=} 0$$

$$\dot{m}_1 s_1 - \dot{m}_2 s_2 + \dot{m}_3 s_3 - \dot{m}_4 s_4 + \dot{S}_{\text{gen}} = 0$$

$$\dot{m}_s (s_1 - s_2) + \dot{m}_{fw} (s_3 - s_4) + \dot{S}_{\text{gen}} = 0$$

$$\begin{aligned} \frac{\dot{S}_{\text{gen}}}{\dot{m}_{fw}} &= \frac{\dot{m}_s}{\dot{m}_{fw}} (s_2 - s_1) + (s_4 - s_3) = (0.247)(2.1381 - 6.6956) + (2.0417 - 0.7038) \\ &= \mathbf{0.213 \text{ kJ/K per kg of feedwater}} \end{aligned}$$

7-193 During a heat transfer process, the entropy change of incompressible substances, such as liquid water, can be determined from $\Delta S = mc_{\text{avg}} \ln(T_2/T_1)$. Show that for thermal energy reservoirs, such as large lakes, this relation reduces to $\Delta S = Q/T$.

Analysis Consider a thermal energy reservoir of mass m , specific heat c , and initial temperature T_1 . Now heat, in the amount of Q , is transferred to this reservoir. The first law and the entropy change relations for this reservoir can be written as

$$Q = mc(T_2 - T_1) \longrightarrow mc = \frac{Q}{T_2 - T_1}$$

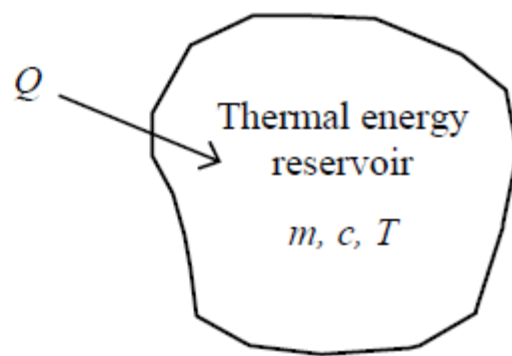
and

$$\Delta S = mc \ln \frac{T_2}{T_1} = Q \frac{\ln(T_2 / T_1)}{T_2 - T_1}$$

Taking the limit as $T_2 \rightarrow T_1$ by applying the L'Hospital's rule,

$$\Delta S = Q \frac{1/T_1}{1} = \frac{Q}{T_1}$$

which is the desired result.



7-196 A passive solar house that is losing heat to the outdoors at 3°C at an average rate of $50,000\text{ kJ/h}$ is maintained at 22°C at all times during a winter night for 10 h. The house is to be heated by 50 glass containers, each containing 20 L of water that is heated to 80°C during the day by absorbing solar energy. A thermostat controlled 15 kW backup electric resistance heater turns on whenever necessary to keep the house at 22°C . Determine how long the electric heating system was on that night and the amount of entropy generated during the night.

7-199 A $4\text{-m} \times 5\text{-m} \times 7\text{-m}$ well-sealed room is to be heated by 1500 kg of liquid water contained in a tank that is placed in the room. The room is losing heat to the outside air at 5°C at an average rate of $10,000\text{ kJ/h}$. The room is initially at 20°C and 100 kPa and is maintained at a temperature of 20°C at all times. If the hot water is to meet the heating requirements of this room for a 24-h period, determine (a) the minimum temperature of the water when it is first brought into the room and (b) the entropy generated during a 24-h period. Assume constant specific heats for both air and water at room temperature.

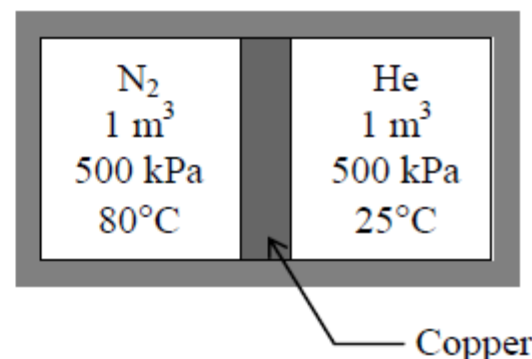
7-200 Consider a well-insulated horizontal rigid cylinder that is divided into two compartments by a piston that is free to move but does not allow either gas to leak into the other side. Initially, one side of the piston contains 1 m^3 of N_2 gas at 500 kPa and 80°C while the other side contains 1 m^3 of He gas at 500 kPa and 25°C . Now thermal equilibrium is established in the cylinder as a result of heat transfer through the piston. Using constant specific heats at room temperature, determine (a) the final equilibrium temperature in the cylinder and (b) the entropy generation during this process. What would your answer be if the piston were not free to move?

Properties The gas constants and the constant volume specific heats are $R = 0.2968 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K}$, $c_v = 0.743 \text{ kJ/kg}\cdot^\circ\text{C}$ and $c_p = 1.039 \text{ kJ/kg}\cdot^\circ\text{C}$ for N_2 , and $R = 2.0769 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K}$, $c_v = 3.1156 \text{ kJ/kg}\cdot^\circ\text{C}$, and $c_p = 5.1926 \text{ kJ/kg}\cdot^\circ\text{C}$ for He (Tables A-1 and A-2). The specific heat of the copper at room temperature is $c = 0.386 \text{ kJ/kg}\cdot^\circ\text{C}$ (Table A-3).

Analysis The mass of each gas in the cylinder is

$$m_{\text{N}_2} = \left(\frac{P_1 V_1}{RT_1} \right)_{\text{N}_2} = \frac{(500 \text{ kPa})(1 \text{ m}^3)}{(0.2968 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(353 \text{ K})} = 4.77 \text{ kg}$$

$$m_{\text{He}} = \left(\frac{P_1 V_1}{RT_1} \right)_{\text{He}} = \frac{(500 \text{ kPa})(1 \text{ m}^3)}{(2.0769 \text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(298 \text{ K})} = 0.808 \text{ kg}$$



Taking the entire contents of the cylinder as our system, the 1st law relation can be written as

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{Net energy transfer by heat, work, and mass}} = \underbrace{\Delta E_{\text{system}}}_{\text{Change in internal, kinetic, potential, etc. energies}}$$

$$0 = \Delta U = (\Delta U)_{\text{N}_2} + (\Delta U)_{\text{He}} + (\Delta U)_{\text{Cu}}$$

$$0 = [mc_v(T_2 - T_1)]_{\text{N}_2} + [mc_v(T_2 - T_1)]_{\text{He}} + [mc(T_2 - T_1)]_{\text{Cu}}$$



where

$$T_{1, \text{Cu}} = (80 + 25) / 2 = 52.5^\circ\text{C}$$

Substituting,

$$(4.77 \text{ kg})(0.743 \text{ kJ/kg} \cdot ^\circ\text{C})(T_f - 80)^\circ\text{C} + (0.808 \text{ kg})(3.1156 \text{ kJ/kg} \cdot ^\circ\text{C})(T_f - 25)^\circ\text{C} \\ + (5.0 \text{ kg})(0.386 \text{ kJ/kg} \cdot ^\circ\text{C})(T_f - 52.5)^\circ\text{C} = 0$$

It gives

$$T_f = 56.0^\circ\text{C}$$

where T_f is the final equilibrium temperature in the cylinder.

The answer would be the **same** if the piston were not free to move since it would effect only pressure, and not the specific heats.

(b) We take the entire cylinder as our system, which is a closed system. Noting that the cylinder is well-insulated and thus there is no heat transfer, the entropy balance for this closed system can be expressed as

$$\underbrace{S_{\text{in}} - S_{\text{out}}}_{\text{Net entropy transfer by heat and mass}} + \underbrace{S_{\text{gen}}}_{\text{Entropy generation}} = \underbrace{\Delta S_{\text{system}}}_{\text{Change in entropy}}$$
$$0 + S_{\text{gen}} = \Delta S_{\text{N}_2} + \Delta S_{\text{He}} + \Delta S_{\text{piston}}$$

But first we determine the final pressure in the cylinder:



$$N_{\text{total}} = N_{\text{N}_2} + N_{\text{He}} = \left(\frac{m}{M}\right)_{\text{N}_2} + \left(\frac{m}{M}\right)_{\text{He}} = \frac{4.77 \text{ kg}}{28 \text{ kg/kmol}} + \frac{0.808 \text{ kg}}{4 \text{ kg/kmol}} = 0.372 \text{ kmol}$$

$$P_2 = \frac{N_{\text{total}} R_u T}{V_{\text{total}}} = \frac{(0.372 \text{ kmol})(8.314 \text{ kPa} \cdot \text{m}^3/\text{kmol} \cdot \text{K})(329 \text{ K})}{2 \text{ m}^3} = 508.8 \text{ kPa}$$

Then,

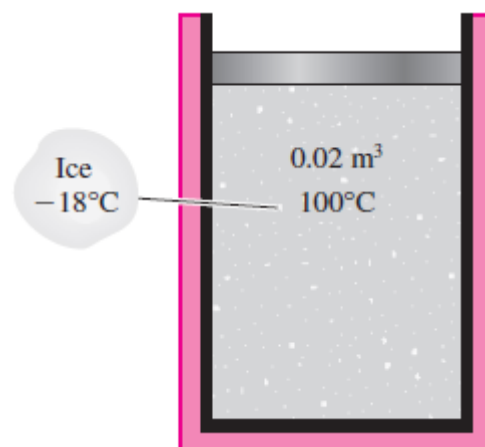
$$\begin{aligned} \Delta S_{\text{N}_2} &= m \left(c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \right)_{\text{N}_2} \\ &= (4.77 \text{ kg}) \left[(1.039 \text{ kJ/kg} \cdot \text{K}) \ln \frac{329 \text{ K}}{353 \text{ K}} - (0.2968 \text{ kJ/kg} \cdot \text{K}) \ln \frac{508.8 \text{ kPa}}{500 \text{ kPa}} \right] \\ &= -0.374 \text{ kJ/K} \\ \Delta S_{\text{He}} &= m \left(c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1} \right)_{\text{He}} \\ &= (0.808 \text{ kg}) \left[(5.1926 \text{ kJ/kg} \cdot \text{K}) \ln \frac{329 \text{ K}}{298 \text{ K}} - (2.0769 \text{ kJ/kg} \cdot \text{K}) \ln \frac{508.8 \text{ kPa}}{500 \text{ kPa}} \right] \\ &= 0.386 \text{ kJ/K} \\ \Delta S_{\text{piston}} &= \left(mc \ln \frac{T_2}{T_1} \right)_{\text{piston}} = (5 \text{ kg})(0.386 \text{ kJ/kg} \cdot \text{K}) \ln \frac{329 \text{ K}}{325.5 \text{ K}} = 0.021 \text{ kJ/K} \\ S_{\text{gen}} &= \Delta S_{\text{N}_2} + \Delta S_{\text{He}} + \Delta S_{\text{piston}} = -0.374 + 0.386 + 0.021 = \mathbf{0.033 \text{ kJ/K}} \end{aligned}$$

If the piston were not free to move, we would still have $T_2 = 329 \text{ K}$ but the volume of each gas would remain constant in this case:

$$\begin{aligned} \Delta S_{\text{N}_2} &= m \left(c_v \ln \frac{T_2}{T_1} - R \ln \frac{V_2}{V_1} \right)_{\text{N}_2} = (4.77 \text{ kg})(0.743 \text{ kJ/kg} \cdot \text{K}) \ln \frac{329 \text{ K}}{353 \text{ K}} = -0.250 \text{ kJ/K} \\ \Delta S_{\text{He}} &= m \left(c_v \ln \frac{T_2}{T_1} - R \ln \frac{V_2}{V_1} \right)_{\text{He}} = (0.808 \text{ kg})(3.1156 \text{ kJ/kg} \cdot \text{K}) \ln \frac{329 \text{ K}}{298 \text{ K}} = 0.249 \text{ kJ/K} \\ S_{\text{gen}} &= \Delta S_{\text{N}_2} + \Delta S_{\text{He}} + \Delta S_{\text{piston}} = -0.250 + 0.249 + 0.021 = \mathbf{0.020 \text{ kJ/K}} \end{aligned}$$

7-203 An insulated 5-m³ rigid tank contains air at 500 kPa and 57°C. A valve connected to the tank is now opened, and air is allowed to escape until the pressure inside drops to 200 kPa. The air temperature during this process is maintained constant by an electric resistance heater placed in the tank. Determine (a) the electrical energy supplied during this process and (b) the total entropy change. *Answers: (a) 1501 kJ, (b) 4.40 kJ/K*

7-205 An insulated piston–cylinder device initially contains 0.02 m³ of saturated liquid–vapor mixture of water with a quality of 0.1 at 100°C. Now some ice at −18°C is dropped into the cylinder. If the cylinder contains saturated liquid at 100°C when thermal equilibrium is established, determine (a) the amount of ice added and (b) the entropy generation during this process. The melting temperature and the heat of fusion of ice at atmospheric pressure are 0°C and 333.7 kJ/kg.



7-206 Consider a 5-L evacuated rigid bottle that is surrounded by the atmosphere at 100 kPa and 17°C. A valve at the neck of the bottle is now opened and the atmospheric air is allowed to flow into the bottle. The air trapped in the bottle eventually reaches thermal equilibrium with the atmosphere as a result of heat transfer through the wall of the bottle. The valve remains open during the process so that the trapped air also reaches mechanical equilibrium with the atmosphere. Determine the net heat transfer through the wall of the bottle and the entropy generation during this filling process.

Answers: 0.5 kJ, 0.0017 kJ/K

Analysis We take the bottle as the system, which is a control volume since mass crosses the boundary. Noting that the microscopic energies of flowing and nonflowing fluids are represented by enthalpy h and internal energy u , respectively, the mass and energy balances for this uniform-flow system can be expressed as

$$\text{Mass balance: } m_{\text{in}} - m_{\text{out}} = \Delta m_{\text{system}} \rightarrow m_i = m_2 \quad (\text{since } m_{\text{out}} = m_{\text{initial}} = 0)$$

Energy balance:

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\substack{\text{Net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\Delta E_{\text{system}}}_{\substack{\text{Change in internal, kinetic,} \\ \text{potential, etc. energies}}}$$

$$Q_{\text{in}} + m_i h_i = m_2 u_2 \quad (\text{since } W \cong E_{\text{out}} = E_{\text{initial}} = \text{ke} \cong \text{pe} \cong 0)$$

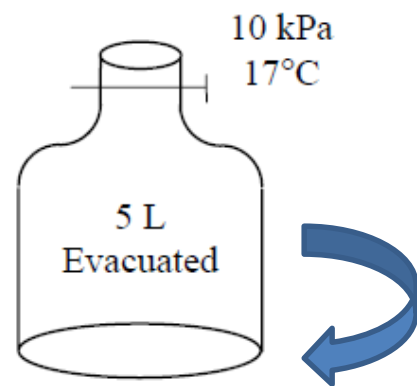
Combining the two balances:

$$Q_{\text{in}} = m_2 (u_2 - h_i)$$

where

$$m_2 = \frac{P_2 V}{RT_2} = \frac{(100 \text{ kPa})(0.005 \text{ m}^3)}{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(290 \text{ K})} = 0.0060 \text{ kg}$$

$$T_i = T_2 = 290 \text{ K} \xrightarrow{\text{Table A-17}} \begin{aligned} h_i &= 290.16 \text{ kJ/kg} \\ u_2 &= 206.91 \text{ kJ/kg} \end{aligned}$$



Substituting,

$$Q_{\text{in}} = (0.0060 \text{ kg})(206.91 - 290.16) \text{ kJ/kg} = -0.5 \text{ kJ} \rightarrow Q_{\text{out}} = \mathbf{0.5 \text{ kJ}}$$

Note that the negative sign for heat transfer indicates that the assumed direction is wrong. Therefore, we reverse the direction.

The entropy generated during this process is determined by applying the entropy balance on an *extended system* that includes the bottle and its immediate surroundings so that the boundary temperature of the extended system is the temperature of the surroundings at all times. The entropy balance for it can be expressed as

$$\underbrace{S_{\text{in}} - S_{\text{out}}}_{\text{Net entropy transfer by heat and mass}} + \underbrace{S_{\text{gen}}}_{\text{Entropy generation}} = \underbrace{\Delta S_{\text{system}}}_{\text{Change in entropy}}$$

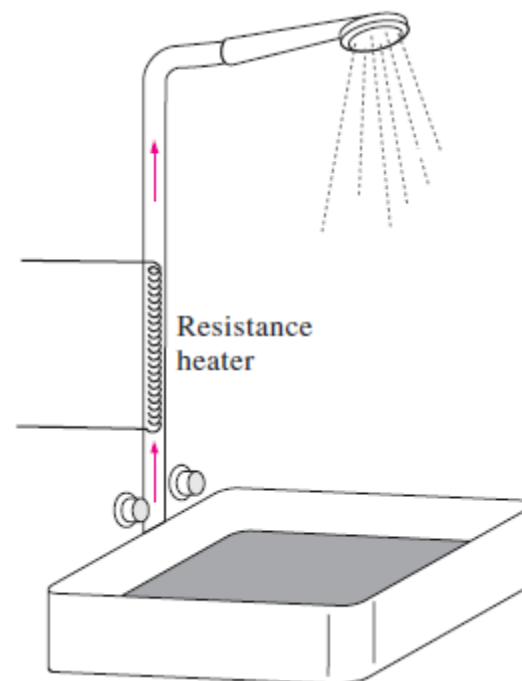
$$m_i s_i - \frac{Q_{\text{out}}}{T_{\text{b,in}}} + S_{\text{gen}} = \Delta S_{\text{tank}} = m_2 s_2 - m_1 s_1 \stackrel{\text{0}}{=} m_2 s_2$$

Therefore, the total entropy generated during this process is

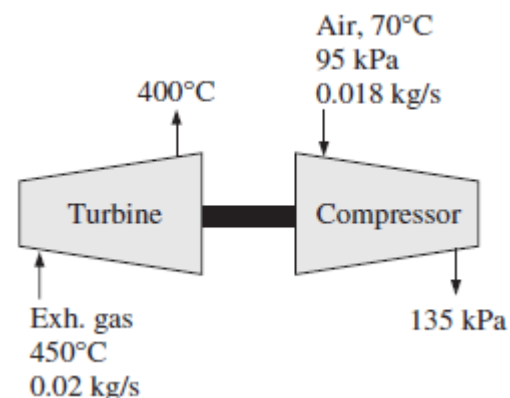
$$S_{\text{gen}} = -m_i s_i + m_2 s_2 + \frac{Q_{\text{out}}}{T_{\text{b,out}}} = m_2 (s_2 - s_i) \stackrel{\text{0}}{=} + \frac{Q_{\text{out}}}{T_{\text{b,out}}} = \frac{Q_{\text{out}}}{T_{\text{surr}}} = \frac{0.5 \text{ kJ}}{290 \text{ K}} = \mathbf{0.0017 \text{ kJ/K}}$$

7-207 (a) Water flows through a shower head steadily at a rate of 10 L/min. An electric resistance heater placed in the water pipe heats the water from 16 to 43°C. Taking the density of water to be 1 kg/L, determine the electric power input to the heater, in kW, and the rate of entropy generation during this process, in kW/K.

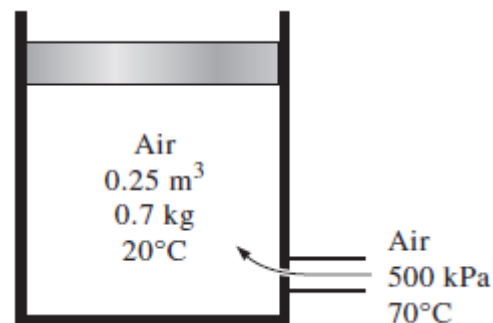
(b) In an effort to conserve energy, it is proposed to pass the drained warm water at a temperature of 39°C through a heat exchanger to preheat the incoming cold water. If the heat exchanger has an effectiveness of 0.50 (that is, it recovers only half of the energy that can possibly be transferred from the drained water to incoming cold water), determine the electric power input required in this case and the reduction in the rate of entropy generation in the resistance heating section.



7-210 Consider the turbocharger of an internal combustion engine. The exhaust gases enter the turbine at 450°C at a rate of 0.02 kg/s and leave at 400°C. Air enters the compressor at 70°C and 95 kPa at a rate of 0.018 kg/s and leaves at 135 kPa. The mechanical efficiency between the turbine and the compressor is 95 percent (5 percent of turbine work is lost during its transmission to the compressor). Using air properties for the exhaust gases, determine (a) the air temperature at the compressor exit and (b) the isentropic efficiency of the compressor. *Answers: (a) 126.1°C, (b) 0.642*



7-212 A 0.25-m^3 insulated piston–cylinder device initially contains 0.7 kg of air at 20°C . At this state, the piston is free to move. Now air at 500 kPa and 70°C is allowed to enter the cylinder from a supply line until the volume increases by 50 percent. Using constant specific heats at room temperature, determine (a) the final temperature, (b) the amount of mass that has entered, (c) the work done, and (d) the entropy generation.



Properties The gas constant of air is $R = 0.287\text{ kJ/kg}\cdot\text{K}$ and the specific heats of air at room temperature are $c_p = 1.005\text{ kJ/kg}\cdot\text{K}$, $c_v = 0.718\text{ kJ/kg}\cdot\text{K}$ (Table A-2).

Analysis The initial pressure in the cylinder is

$$P_1 = \frac{m_1 R T_1}{V_1} = \frac{(0.7\text{ kg})(0.287\text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})(20 + 273\text{ K})}{0.25\text{ m}^3} = 235.5\text{ kPa}$$

$$m_2 = \frac{P_2 V_2}{R T_2} = \frac{(235.5\text{ kPa})(1.5 \times 0.25\text{ m}^3)}{(0.287\text{ kPa}\cdot\text{m}^3/\text{kg}\cdot\text{K})T_2} = \frac{307.71}{T_2}$$

A mass balance on the system gives the expression for the mass entering the cylinder

$$m_i = m_2 - m_1 = \frac{307.71}{T_2} - 0.7$$

(c) Noting that the pressure remains constant, the boundary work is determined to be

$$W_{b,\text{out}} = P_1 (V_2 - V_1) = (235.5\text{ kPa})(1.5 \times 0.25 - 0.25)\text{ m}^3 = \mathbf{29.43\text{ kJ}}$$

(a) An energy balance on the system may be used to determine the final temperature

$$m_i h_i - W_{b,out} = m_2 u_2 - m_1 u_1$$

$$m_i c_p T_i - W_{b,out} = m_2 c_v T_2 - m_1 c_v T_1$$

$$\left(\frac{307.71}{T_2} - 0.7 \right) (1.005)(70 + 273) - 29.43 = \left(\frac{307.71}{T_2} \right) (0.718) T_2 - (0.7)(0.718)(20 + 273)$$

There is only one unknown, which is the final temperature. By a trial-error approach or using EES, we find

$$T_2 = \mathbf{308.0 \text{ K}}$$

(b) The final mass and the amount of mass that has entered are

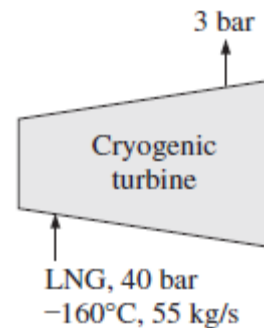
$$m_2 = \frac{307.71}{308.0} = 0.999 \text{ kg}$$

$$m_i = m_2 - m_1 = 0.999 - 0.7 = \mathbf{0.299 \text{ kg}}$$

(d) The rate of entropy generation is determined from

$$\begin{aligned} S_{\text{gen}} &= m_2 s_2 - m_1 s_1 - m_i s_i = m_2 s_2 - m_1 s_1 - (m_2 - m_1) s_i = m_2 (s_2 - s_i) - m_1 (s_1 - s_i) \\ &= m_2 \left(c_p \ln \frac{T_2}{T_i} - R \ln \frac{P_2}{P_i} \right) - m_1 \left(c_p \ln \frac{T_1}{T_i} - R \ln \frac{P_1}{P_i} \right) \\ &= (0.999 \text{ kg}) \left[(1.005 \text{ kJ/kg.K}) \ln \left(\frac{308 \text{ K}}{343 \text{ K}} \right) - (0.287 \text{ kJ/kg.K}) \ln \left(\frac{235.5 \text{ kPa}}{500 \text{ kPa}} \right) \right] \\ &\quad - (0.7 \text{ kg}) \left[(1.005 \text{ kJ/kg.K}) \ln \left(\frac{293 \text{ K}}{343 \text{ K}} \right) - (0.287 \text{ kJ/kg.K}) \ln \left(\frac{235.5 \text{ kPa}}{500 \text{ kPa}} \right) \right] \\ &= \mathbf{0.0673 \text{ kJ/K}} \end{aligned}$$

7-213 When the transportation of natural gas in a pipeline is not feasible for economic reasons, it is first liquefied using nonconventional refrigeration techniques and then transported in super-insulated tanks. In a natural gas liquefaction plant, the liquefied natural gas (LNG) enters a cryogenic turbine at 40 bar and -160°C at a rate of 55 kg/s and leaves at 3 bar. If 350 kW power is produced by the turbine, determine the efficiency of the turbine. Take the density of LNG to be 423.8 kg/m^3 . *Answer: 72.9 percent*



Chapter 7

End of the Problems